

**ANALYSIS OF THE IMPACT OF SELECTED MACROECONOMIC
VARIABLES ON INFANT MORTALITY IN NIGERIA (1990-2024).**

BY

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BHU/23/MSC/ECO/028**

MASTER OF SCIENCES (M.Sc.) ECONOMICS

M.Sc. Dissertation, 2025



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ON INFANT MORTALITY IN NIGERIA (1990-2024).**

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**A DISSERTATION SUBMITTED TO THE SCHOOL OF POSTGRADUATE
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SCIENCE (M.Sc.) DEGREE IN ECONOMICS**

**DEPARTMENT OF ECONOMICS
FACULTY OF SOCIAL SCIENCES
BINGHAM UNIVERSITY KARU
NASARAWA STATE**

DECLARATION

I hereby declare that; this dissertation proposal titled “**Analysis of The Impact of Selected Macroeconomic Variables on Infant Mortality in Nigeria: (1990-2024)**” has been written by me and it is a report of my research work. It has not been presented in any previous application for Masters of Science (M.Sc.) Degree in Economics. All quotations are indicated and sources of information specially acknowledged by means of references.

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AUGUSTINE IFEANYI CYRIACUS
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CERTIFICATION

This dissertation entitled “**Analysis of the impact of selected macroeconomic variables on infant mortality in Nigeria: (1990-2024)**” meets the regulations governing the award of (Master of Science degree in Economics), of the school of Postgraduate Studies of Bingham University, Karu for its contribution to knowledge and literary presentation.

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DEDICATION

To my beloved wife, for your utmost patience, encouragement, and believe with faith in my dreams. This dissertation paper is as much yours as it is mine.

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ABSTRACT

Good health is indispensable for the productivity of human capital and remains an important factor for the realization of lasting economic growth and development. This study empirically investigates the impact of selected macroeconomic variables on infant mortality in Nigeria, with particular emphasis on access to electricity as a crucial indicator of health outcomes. Specifically, the study examines the impacts of Gross Domestic Product per Capita (GDPPC), Government Health Expenditure (GHE), Access to Electricity (ACE), and Health-Specific Aid (AID) on infant mortality using the Autoregressive Distributed Lag (ARDL) technique on annual data from 1990–2024. For the long-term findings, access to electricity was seen to significantly reduce infant mortality as the value of the coefficient came up as -2.048302 and a p -value of 0.0018 . This pointed to an impending health implication that may occur given the performance of the energy sector, greatly dependent on how monetary and fiscal policies decide the level of investments in essential energy infrastructure. Also, health-specific aid had a strong, negative, and statistically significant impact reflected by the coefficient of -0.059087 and probability of 0.0009 , emphasizing the critical role of sufficient external funding to complement domestic resources. On the contrary, GDP per capita shows a weak and statistically insignificant long-term impact as the coefficient came out as 0.002508 with p value of 0.0956 , indicating that poor monetary policy stance has resulted to poor public health. Similarly, government health expenditure is found to be insignificant in the long run given the p value of 0.7099 , a reflection of inefficiencies in the fiscal policy and inadequate budget execution within the health sector. The results in the brief period further emphasized the importance of electricity access. The analysis reveals enough negative and significant lags related to ACE, while the error-correction term of -1.06046 with its corresponding p value of 0.0000 confirms a stable long-run equilibrium exists, although the convergence to stability happened slowly. Generally, the findings are grounded on the necessity of expanding electricity access, improving the quality and reliability of supply, and ensuring efficiency in the administration of public sector to effectively transform macroeconomic policies into better health expectations. The inquiry recommends a steady and dependable macroeconomic policy, investments in renewable energy, and attracting health-specific aid as indispensable strategies for reducing infant mortality and staying on track of attaining sustainable economic development goals in Nigeria.

CHAPTER ONE

INTRODUCTION

1.1 Background to the study

Nigeria is not developed yet, for this reason, the health care system should be made to be like those in developed countries, because it will curb the issue of premature infant demise and as well cause the nation to grow in the right direction. Improved health outcomes enhance productivity, alleviate the financial burden of medical care, and strengthen human capital. In simple terms, a healthier population is more productive, improves workforce quality, and stimulates economic activity (World Health Organization, 2021). According to Aigbedion (2017), allocating enough funds towards the well-being of the citizens is a step in the right direction. When this is done, the rate of newborn deaths is expected to drop as a result of corresponding economic growth and development. For Nigeria, sufficient funding and investment should be taken seriously while seeing them as an indispensable tool for achieving far-reaching result.

Human capital is a crucial element for the attainment of everlasting growth and development in Africa, particularly in Nigeria where the rate of new born deaths is on the rise. Nigeria's Federal health authority, Ministry of Health, on 8 September in the year 2000 has responded to this threat through health policies and pairing health strategy with the Millennium Development Goals while joining 189 other United Nations member states in committing to address global challenges, which includes child mortality. This commitment provided a framework for monitoring progress and ensuring accountability. Nigeria now is on track to achieving a reduced infant death to a reasonable amount by the year 2035 (Federal Ministry of Health, 2018).

1 The Nigerian state is very intentional on health-related matters, for example, when the world
2 changed from millennium Development Goals (MDGs) to the Sustainable Development Goals
3 (SDGs) along with other members of United Nations, the nation also adjusted policies to match
4 this change (Child Health Task Force 2019). Additionally, these policies were designed in way
5 that it provides the presiding Authority and their partners the platform required to keeping funds
6 allocation and human resources in check, while ensuring their targets of improving the health
7 of newborns and children are met.

8

9 United Nations (2015), explained that, to reach a low figure of newborn deaths, it is necessary
10 to couple health policies and strategies with policies that focus economic growth, national
11 equity, and environment, because national healthiness and well-being is in proximity with
12 sustainable development strategies of the 2030 Agenda. Furthermore, Agenda 2030 entails:
13 reaching a global health coverage, reducing children's deaths, and reducing contagious
14 sicknesses. This Agenda is connected to Brundtland Commission formally also referred to as
15 World Commission on Environment and Development, chaired by Gro Harlem Brundtland. By
16 2030, the SDGs are expected to be fully achieved, provided there is consistent commitment
17 from all stakeholders. However, progress differs widely across countries.

18

19 For example, World Bank (2025) data on infant mortality shows that Norway, through its robust
20 healthcare and social welfare systems, maintains very low child death rates of about 2–3 per
21 1,000 new deliveries. In contrast, Nigeria still records between 55 to 75 demises compared to a
22 thousand new births, reflecting the impact of inadequate healthcare infrastructure, widespread
23 poverty, and persistent governance challenges. Technology is very important in strengthening

1 health systems, standardizing medical facilities, and improving outcomes such as reducing
2 infant mortality. Solow (1956), in a theoretical exposition, provided a comprehensive model to
3 justify this connection. The model pointed to capital accumulation, labor, and technology as the
4 major elements that ensures lasting economic growth and development. Among these,
5 technological progress is emphasized as the most critical factor in sustaining development.
6 Applied to healthcare, this perspective highlights how innovations and access to enabling
7 infrastructure such as reliable electricity directly affect the efficiency and quality of medical
8 services. Together, these variables form a sensible framework for understanding how
9 macroeconomic indicators shape infant mortality in line with the growth dynamics described
10 by Solow (1956). Idris & Abdullahi (2024) in a recent empirical work show that increased
11 public investment in child health interventions statistically relate with lower infant mortality in
12 Nigeria over the period 1995–2023, reinforcing the relevance of combining macroeconomic
13 and health related variables in this research.

14
15 In the Nigerian case, insufficient access to modern medical technology and unstable electricity
16 supply have constrained healthcare delivery, with adverse effects on infant survival rates.
17 Notably, investing in technology-driven healthcare solutions and improving energy access can
18 significantly enhance productivity in the health sector by reducing preventable deaths and
19 sustainable economic growth. This is particularly relevant to the role of technology in
20 healthcare and electricity access in Nigeria. In this regard, Ajala (2022) highlighted that
21 technology is a missing-link in driving improvements on factors that could lead to infant
22 mortalities in Nigeria, emphasizing on the impact of electricity access and economic policies
23 that support sustainable growth. He further noted, that technology can address the problem of

1 high infant death in the nation by maintaining a global standard healthcare access, efficient
2 health services, reliable energy crisis solutions, and addressing environmental pitfalls. It is a
3 no-brainer that the role of government is indispensable in growing an economy while pairing it
4 with improved health outcomes such as infant mortality rate geared towards an inclusive
5 sustainable growth and development. This is achieved through the formulation of health-
6 supportive economic policies, investment in public health infrastructure, regulation of industry
7 and environment, ensuring universal health coverage, public health education and awareness,
8 creating avenues for foreign partnership and investments, monitoring and accountability in
9 governance. Government through these can directly impact the ability of a nation to ensure
10 reduced infant deaths and equitable health outcomes for all citizens (Vu *et al.*, 2024).

11

12 Health is a major factor in the survival of the human race and cannot be underestimated.
13 Therefore, it is essential and requires all hands-on deck to ensure access to good and quality
14 health facilities and to promote and develop the health sector into a sort after world-class
15 category. Health is fundamental to the survival and progress of humanity and cannot be
16 underestimated. Ensuring access to quality healthcare services, therefore, requires a collective
17 commitment to building and strengthening health systems that meet global standards. The
18 authority responsible for health-related matters in Nigeria is the Federal Ministry of Health
19 (FMoH), the organization ensures that the system operations run smoothly, efficiently, and
20 effectively in accordance with blueprint for the nation's healthcare delivery system. The
21 government of Nigeria through this ministry ensures that the financial facilities made available
22 are purposely used for national health plans that targets general good health and well-being of
23 the citizens (Abubakar *et al.*, 2022). Furthermore, the Nigerian government is the architect in

1 deciding the pattern with which the national health plan is implemented bearing over 90%
2 financial burden.

3 Similarly, state governments align with this structure, drawing influence from federal spending
4 while granting local governments some level of autonomy in resource control. Through their
5 respective ministries, state governments are responsible for harmonizing state-level health
6 programs with national priorities, as well as managing state health institutions. At the grassroots
7 level, local governments also take on the role of ensuring equitable access to healthcare services
8 within their jurisdictions (Vu *et al.*, 2024).

9

10 **1.2 Statement of the Problem**

11 According to UNICEF (2019), Nigeria accounted for approximately 28% of infant deaths in
12 sub-Saharan Africa, with 548,116 cases out of a total of 1.9 million infant deaths in the region.
13 In contrast, countries like Kenya, Ghana, and South Africa boasted of significantly lower infant
14 death rates, at 31, 35, and 28 deaths per 1,000 new deliveries. These figures show that while
15 these sub-Saharan African nations are on the right lane towards reaching the objectives of the
16 SDGs which is set to end in the year 2030, Nigeria continues to struggle with alarmingly high
17 infant mortality rates. This gap raises significant cause for concerns about the actual reason
18 behind the rising deaths of newborns in in Nigeria and questions the effectiveness of current
19 policies and interventions put out to put an end to the problem.

20

21 Nigeria's healthcare sector is in a sorry state and there is nothing good to write home about it.
22 This is attributed to the impact of the past bad governance- and the present-day Nigerian
23 economy and this has negatively impacted health outcomes such Infant Mortality Rate (IMR)

1 which can be described as potentially catastrophic. It is nothing but shameful that as the so-
2 called Giant of Africa with regards to humongous population boosting the human capital with
3 spill over that are also making waves in other countries and the numerous natural resources that
4 can potentially turn the economy around, but unfortunately the healthcare system has been
5 affected negatively despite the potentials of Nigeria's economy and by this has suffered a
6 significant neglect (Nkem, 2021). Furthermore, inadequate primary healthcare facilities,
7 insufficient professional health workers, and epileptic power supply are the major factors
8 leading to reduced infant mortality across Nigeria. Unfortunately, there has not been significant
9 progress in achieving key health sustainable development (Okereke *et al.*, 2019). OPEC (2023)
10 noted that in the crude oil business globally, Nigeria sits among top producers while standing
11 as the highest producer in Africa. But disappointingly, the country's effort towards allocating
12 the sufficient funds needed to transform the current sorry state of the health delivery remains
13 among the lowest across the globe. In the 2024 national budget, the federal government
14 allocated ₦1.33 trillion to the health sector representing only 5% of the total budget, far below
15 the 15% commitment endorsed by African Union member states under the Abuja Declaration
16 (Simon, 2024). This recurring underfunding has left many Nigerians' struggling to access
17 formal healthcare services. The consequences are seen in health indicators such as infant
18 mortality, which have remained stubbornly high despite successive policy initiatives. Beyond
19 falling short of funding, the sector faces other structural problems such as weak leadership,
20 limited investment in research and technology, fragmented service delivery, inadequate
21 financing mechanisms, and poor health infrastructure (Bakare, 2021). Collectively, these
22 challenges undermine the effectiveness of health interventions and contribute to Nigeria's slow
23 progress in reducing infant mortality compared to its peers.

Nigeria has paid its dues since the millennium declaration of 8th September, 2000 and has committed to various health policies coupled with macroeconomic reinforcing interventions. These efforts have proved futile, because newborn demise is still on the rise and never ending as against what is expected internationally. Some economic macroeconomic variables like health care expenditure, income per capita, inflation, and rate of poor people in the country have shown to influence health outcomes, especially infant mortality. For example, Nigeria have focused on stabilizing this variable through targeted health programs and increasing public investment. This have led to a significantly reduced demise of newborns from 112 deaths per 1,000 new births in 2000 to approximately 55–60 by the year 2023 (Macrotrends, 2024). While this progress is praiseworthy, the rate still exceeds global averages and those of comparable African nations. This hopeless disparity highlights the urgent need to explore how various macroeconomic factors such as income growth, health spending, access to electricity, and external aid interact to influence infant mortality outcomes in Nigeria. This study on that note carried out a critical observation on macroeconomic factors such as - income per capita, government health expenditure, access to electricity, an indicator and proxy for technology advancement, and health-specific aid.

1.3 Research Questions

The following questions will be addressed in this study:

- i. How does per capita income shape infant mortality in Nigeria?
- ii. How does spending on health by relevant authority impact infant mortality in Nigeria?
- iii. What impact does access to reliable electricity infrastructure have on infant mortality in Nigeria?

- iv. What is the role of health-specific aid in reducing infant mortality in Nigeria?

1.4 Objectives of the Study

The main purpose behind this study is to explore how certain macroeconomic agents influence infant mortality in Nigeria. Secondly, this research looks to:

- i. Assess the influence of income per capita on infant mortality in Nigeria.
- ii. Examine the impact of health spending by the relevant authority on infant mortality in the country.
- iii. Analyze the role of access to reliable electricity infrastructure in influencing infant mortality in Nigeria.
- iv. Evaluate the impact of health-specific aid on infant mortality in Nigeria.

1.5 Statement of Hypotheses

The following hypotheses are proposed to be tested.

H₀₁: There is no significant impact of GDP per capita on infant mortality in Nigeria.

H₀₂: Government health expenditure has no significant impact on infant mortality in Nigeria.

H₀₃: There is no significant impact of access to electricity on infant mortality.

H₀₄: Health-specific aid has no significant impact on infant mortality.

1.6 Significance of the study

The outcome as established by this study will be very useful for policymaking, development planning, and health care practitioners in equipping targeted strategies aimed at reducing IMR.

Also, it will enrich existing academic literature by bridging the gap between macroeconomic

performance and child health outcomes while highlighting the importance of non-traditional factors such as electricity access in shaping health indicators. By establishing the relationship that exists between these indicators and how they influence infant mortality, the study emphasizes the importance of investing in healthcare systems, policies, and relevant programs. This research through relevant comparison between Nigeria's IMR trends with other well performing economies of the sub-Saharan region for example south Africa, Ghana, and Senegal can inform policymakers by highlighting successful health and economic policies that can be adapted by healthcare providers, major stakeholders, and businesses about the long-term economic benefits of improving health standards, such as reducing healthcare costs, enhancing workforce efficiency, and fostering sustainable development. The result gotten from this investigation will also contribute to the growing literature that seek to discover how certain macroeconomic indicators influence infant mortality in developing countries. By offering data-based recommendations, it aims to support Nigeria in improving its performance toward achieving the Sustainable Development Goals (SDGs). Apart from its policy relevance, the outcome of this investigation will be of great assist to researchers, scholars, and students particularly within the Department of Economics at Bingham University by serving as a useful reference for academic inquiry. It seeks to bridge existing gaps in the literature while also providing a foundation for future research on health-related problems like persistence demise of new born babies.

1.7 Scope of the Study

The idea driving this research roots from the passion to contribute to the numerous sources of ideas to the relevant health authorities in their quest to formulate policies, targeted at desired

1 economic development output to relatively reduce the rate of infant mortalities in Nigeria. It
2 prescribes possible way through which these authorities can manipulate the necessary
3 macroeconomic indicators such as income per capita, government health expenditure, health-
4 specific aid, and by Improving access to electricity and many other macroeconomic factors to
5 deliver a lasting solution. What separate this research from others is that it creates a scenario
6 where improvement in economic growth and development coupled with focus on providing a
7 lasting solution to the energy crises that bedevil the power sector leads to a lower infant
8 mortality rate and a sustained good child health condition in Nigeria. This study carried out
9 examination of this issue with the data from gotten from the years 1990 to 2024, and these years
10 observed stands to give this research a firm result that will undoubtedly help in redirecting the
11 government, key stakeholders to power, and health sectors which will in turn facilitate low
12 infant mortality in Nigeria.

14 **1.8 Organization of the Study**

15 In a Sequent pattern, the inquiry into these elements of the economy in general and how it
16 decides the faith of newly delivered babies in the state of Nigeria is phased into five chapters.
17 The first phase, introduces the study, presenting the background, statement of the problem,
18 research questions, objectives, hypotheses, significance, scope, and structure of the research.
19 Second chapter provides a comprehensive literature review, covering the conceptualization of
20 macroeconomic determinants and infant mortality rate, theoretical framework, empirical
21 evidence, and the identified gap this investigation intend to cover. The third phase outlines'
22 methods employed in this investigation, including the design applied, sources of data, model
23 expression, and methods of analysis. The fourth phase delves into the central part of the

1 research, it presents and analyzes data on the selected macroeconomic determinants and how
2 they affect infant death rate in Nigeria. The final phase provides summary to the major
3 outcomes, draws conclusions, prescribes solutions, outlines limitations, and highlights the
4 study's contributions to knowledge.

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CHAPTER TWO

LITERATURE REVIEW

2.1 Conceptual Review

The core variables this investigation focus on are: Infant Mortality Rate (IMR), the explained variable and a set of selected macroeconomic determinants, which serve as the explaining variables. These macroeconomic factors are expected to influence the health outcomes of Nigeria's population, particularly infant survival. Within this framework, the independent variables are further broken down into four sub-components: Gross Domestic Product per Capita (GDPPC), Government Health Expenditure (GHE), Access to Electricity (ACE), and Health-Specific Aid (AID). Each of these indicators is conceptually important as they provide unique link through which economic conditions and public investment can affect child health and, ultimately, infant mortality outcomes in Nigeria.

2.1.1 Macroeconomic Indicator

Macroeconomic indicator is an economic factor, an essential determinant of the health and performance of the economy. These indicators are to be considered when carrying out health care services and during procurement of health care products that may determine the status of national health. For example, the adoption of a certain fiscal based economic policy that targets the health sector could influence the government health expenditure (Naik, *et al.* 2017).

2.1.2 Gross domestic product per capita

In practice, national income per head is derived by dividing GDP by the number of people in that nation (NBS, 2019). Bhowmik, D. (2025) in a study explained that Gross Domestic Product

(GDP) consists of the quantitative cost of all goods and services produced by a country during a financial year, not considering the identity of the people or the actors involved in production. GDP does not account for depreciation and can be measured using different approaches such as expenditure, income, or output. Under the expenditure approach, GDP includes components such as Gross Fixed Capital Formation (investment in fixed assets like buildings and machinery), household final consumption, and government or not-for-profit transfers in kind to households. GDP in general is divided into Nominal GDP (NGDP) and Real GDP (RGDP). As Chanchal (2024) explains, NGDP measures output at current market prices, while RGDP takes into account general price increase or decrease, providing the actual pace of economic growth. This indicator serves as an essential measure of living standards, since a higher GDP per capita means greater purchasing power and improved access to essential goods and services. For developing nations like Nigeria, GDP per capita provides insights into the capacity of citizens to afford healthcare, education, housing, and adequate nutrition. Higher levels of income per person not only improve access to quality health services but also improve human capital outcomes, thereby contributing to reduced infant mortality and broader socio-economic development.

2.1.3 Government health expenditure

The increasing healthcare cost as an issue has become a major concern globally where efforts to control cost has brought about an ongoing debate which looks to address the issues regarding efficiency and sustainability of healthcare spending using value-based care models, preventive care, and health care reforms. According to the World Health Organization (WHO, 2021), international health spending has been rising rapidly, creating challenges for achieving

1 sustainable development and ensuring universal access to quality healthcare services.
2 Healthcare expenditure simply means the sum amount of money value spent on healthcare
3 services and products for a specific period of time, usually per annum. This money spent
4 exclusively comprises the spending by government on amenities, facilities, wages, and other
5 health-related expenses like pharmaceutical products, nursing homecare, and administrative
6 costs. There are factors that influences the spending of healthcare such as – Demographic
7 changes which for example is money spent in other to acquire healthcare services for Aging
8 population; price inflation of all the medical products and services; policies and reform which
9 are drastic actions taken by government such as- health provisions and insurance schemes.

10

11 Based on the information provided by the Centers for Medicare & Medicaid Services (2023),
12 spending on national healthcare constitutes - Public spending: This is mainly the expenditure
13 by governmental bodies at all levels, such as the federal, state, and local. This expenditure spans
14 a wider range of services and programs targeted at ensuring quality public health, availing
15 healthcare essentials to vulnerable populations, and maintaining a solid healthcare
16 infrastructure across the nation; Private spending: This is the expenditure by the private sector,
17 includes personal health spending by ordinary citizens for instance, procuring health insurance
18 and services provided by non-governmental organizations; Other Sources: This form of
19 expenditure includes foreign investments, aids, and philanthropic contributions.

20

21 **2.1.4 Access to electricity**

22 According to Sharma et al. (2024), electricity is essential for the effective functioning of
23 healthcare facilities. It supports the use of medical equipment, ensures safe deliveries, facilitates

1 neonatal care, preserves vaccines, and enables modern health information technologies.
2 Without a steady electricity supply, the quality and efficiency of healthcare services suffer
3 significantly. Ritchie (2021) further highlights that access to electricity is a critical indicator of
4 development, defined as the percentage of a country's population that has a dependable and
5 uninterrupted power supply. Therefore, beyond being a basic amenity, access to electricity is a
6 fundamental factor that play a very vital part in reforming health outcomes and effectively
7 curbing persistent deaths of newborns, especially in developing countries like Nigeria. Also,
8 Aonat (2024), pointed out the importance of technology in the Healthcare system, noting that
9 technological advancement encompasses all forms of innovation in all fields and sectors such
10 as: information technology, energy, and materials and science, et cetera.

11

12 **2.1.5 Health-specific aid**

13 Health aid is the financial support received by a country which assist in the development of its
14 healthcare system. However, this aid for Nigeria is a critical supplement in addition to the
15 government healthcare expenditure. This health aid majorly come from – bilateral and
16 multilateral donors, grants, public-private partnership, NGOs, etc. (Mass life Healthcare
17 Limited, 2025). According to Helgason (2016), health aid has evidently proved to significantly
18 impact the health level of majority of the recipient countries, indicating that well-targeted aid
19 is beneficial by improving the health of the country’s citizens, thereby reduces infant mortality.
20 Also, for health aid to deliver the benefits that it brings, it is highly dependent on how organized
21 the recipient nation can get. This simply implies to the fact that countries with strong institutions
22 which is one characterized by good governance, accountability, efficiency, and effective health
23 systems tend to utilize aid efficiently.

2.1.6 Infant mortality

Health is a broad and complex concept and in every part of the world, healthcare systems vary. Health Outcomes however, looks into understanding and developing a framework for evaluating and assessing the effectiveness of healthcare interventions, environmental effects on an individual or general population's health status, thereby developing quality healthcare delivery. Lee and Leung (2014) define health outcomes as the consequences experienced by patients following their interaction with the healthcare system. Among these, infant mortality remains one of the most critical indicators of a nation's overall health status and the effectiveness of its healthcare delivery. In Nigeria, the rate is alarming, with about 72 infants out of every 1,000 live births not surviving their first year of life (World Bank, 2023). In other words, for every 1,000 babies born, 72 do not make it to their first birthday, reflecting both the fragility of health systems and persistent socio-economic challenges. Infant mortality is derived from subtracting all the demised new born babies that have not attained age 1 from the total of 1000 babies born in that period. Godwin (2017) further illustrates this challenge in a report on the Braithwaite Memorial Specialist Hospital (BMSH) in Rivers State, where a protest by nursing mothers followed the tragic deaths of two premature infants during a power outage. According to a hospital staff, one of the babies died due to problem caused by low temperature and a very low birth weight, while the other, born extremely prematurely, could not survive without constant power-dependent medical support. This is an example of the sorry state of Nigeria's healthcare system and how technology advancement may impact infant mortality outcome.

2.2 Theoretical Review

This research draws on five key theories. They are: Solow-Swan Growth Theory, Grossman Model of Health demand, Modernization Theory, Theory of Human Capital, and Health Led Growth Hypothesis.

2.2.1 Solow-Swan growth theory

The complex design of how an economy grows in exogenous manner, as crafted by Solow and Swan (1956), lay emphasis on the elements and agents that determine the direction of how an economic activity grows to a greater height. The agents highlighted are: high capital growth, labor but largely influenced by the total number of people involved in economic activity, and technological progress, with the latter considered the primary long-term determinant of growth. The model assumes that technological advancement grows exogenously and remains the main determinant of sustained development. In simple terms, the Solow Model explains how an economy expands over decades rather than in short periods. When microeconomic trends are incorporated, the model is often expressed using a Cobb-Douglas production function to address issues of aggregate output (Acemoglu & Daron, 2009).

Frey (2017) notes that the model remains widely applied because it highlights the role of total factor productivity in raising living standards. Later refinements, such as the Ramsey–Cass–Koopmans model, introduced household optimization into the framework, enriching its analysis of growth dynamics. In its simplest form, the model is represented by a Cobb–Douglas production function, which helps economists capture how economies reach equilibrium and expand over time. According to the Corporate Finance Institute (2024), the model often

1 borrowed the Functional form of production, as mathematically crafted by Charles Cobb and
2 Paul Douglas, often applied by neoclassical growth theorists to analyze economic growth and
3 equilibrium in a structured way. Thus, the following equation:

$$4 \quad Y = A \cdot K^{\alpha} \cdot (L)^{1-\alpha} \quad 2.1$$

5 Where: Y is for GDP, A stands for the determinant(s) of technology, K is the capital stock, L
6 stands for labor inputs, and Alpha (α) represents the output elasticity of capital usually standing
7 between zero and one.

8 Health outcomes such as infant mortality can be influenced by capital accumulation for
9 example, the accumulation of wealth could increase government health expenditure and
10 building of infrastructures as well as technological advancement like steady electricity as a
11 result of investment in renewable energy and medical innovations (Makiew *et al.*,1992). This
12 way, a brighter future is assured, meaning that a decline in the infant deaths would lead to
13 sufficient human capital (labor force) and productivity leading to promising and lasting
14 economic performance. This structure was adapted in the inquiry as it fits the variables and
15 would ensure a corresponding expectation at the end of the analysis. It will hypothesize that
16 improvements in electricity access and the other explaining variables contribute to
17 improvements on infant mortality by enhancing the productivity of health services, improving
18 living conditions, and fostering economic growth. the core economics elements screened is this
19 inquiry will borrow the complex mechanism of economic activity growth as crafted by Solow
20 and Swan to give this research smooth sailing and to effectively explore how these
21 macroeconomic factors influence health results in this African state.

22

23

2.2.2 Grossman model of health demand

This particular mechanism was first introduced by Michael Grossman in “The Demand for Health: A Theoretical and Empirical Investigation (1972)”, emphasizes health as a durable good that individuals inherit naturally but which depreciates over time. In this value, medical care and related activities are seen as investments aimed at maintaining or improving health, while natural aging and depreciation represent the gradual decline of health status. According to Jones et al. (2012), the model highlights the dual role of health such as: it is both a consumption good that directly provides satisfaction and a capital good that extends the time available for productive and non-market activities. The demand for health care is thus termed as an optimization problem, where individuals extend their lifetime satisfaction by balancing medical consumption, time devoted to health investments, and the inevitable depreciation of health. notably, the model treats life expectancy as an endogenous element, influenced by individual choices regarding health investment. According to Wagstaff (1986), in an effort to further expand the model developed a simple mechanism that offering more perspective into what was opined by Grossman. Wagstaff in this vein established four basic premises that underlie the theory which is as follows: good health is satisfying but not prioritized among other needs; individual's health depends greatly on their capacity to obtain health sustaining tools like - health care, food, etc.; consumption of the health sustaining tools and other basic life needs yields a positive result for the consumer; and lastly that person's resources are exhaustible. These implies that quasi-concave utility function “U” is present, and is defined by health denoted by “H”, and consumption “C”. Additionally, the next premise emphasizes that the function of health in production applies health inputs “M”, produces health as in the function $H(M)$, where $H_1(\bullet) > 0$ and $H_2(\bullet) < 0$. The extra product of health is an increasing function of

1 the state of technical knowledge: development in medicine is as a result of technical
 2 advancement hence tends to generally increase $H_1(\bullet)$, meanwhile the most educated individuals
 3 have high chance of being well-informed on health rhetorics and thus have a remarkable value
 4 of $H_1(\bullet)$ compared to less-educated. However, the later premises is in form of budget constraint
 5 expressed as:

$$6 \quad P_c C + P_m M = Y \text{-----} \quad 2.2$$

7 Where P_c and P_m stands for each price of unit C and M while Y stands for income. In this case
 8 this person prefers and acquire good C and M to maximize utility subject to the health
 9 production function and the budget constraint. This model avails a strong theoretical framework
 10 for empirical analysis of how macroeconomic indicators impact on infant mortalities given
 11 investments on health sector in Nigeria.

12

13 **2.2.3 Modernization theory**

14 Acemoglu *et.al.* (2022) in a review discovered that this theory encompasses theories that makes
 15 up the modernization theory which holds that the more and more a society become
 16 economically wealthy, healthy, educated and developed, that it may reflect definitely on their
 17 political institutions as they would become liberal democratic and rationalist. Inglehart and
 18 Wenzel (2005), defined modernization theory as “the process by which societies move from
 19 traditional or pre-modern conditions to those of modernity, characterized by industrialization,
 20 urbanization, and the growth of a mass society.” In other words, it means a step-by-step process
 21 through which a nation grows from grassroot level to a higher-level nation. The theory was
 22 crafted in 1950s and 1960s stemming from the ideas of max weber, and was further expanded
 23 by Talcott Parsons. It emphasized that for underdeveloped nations to escape poverty it

prescribes the need to encourage development by investment and technology transfers. However, Prateek (2010), in a study noted that modernization theory investigates into the transition through which economic possibilities and growth leads to development in societies. Also, Rostov (1990), highlighted that industrialization is key to actualizing modernization goals. Frey *et al.* (2016) argued that modernization theory lean toward the idea that industrialization and the economic development that comes after can cause a decline in infant mortality through improvements in healthcare services, education, nutrition, and related areas. Empirical studies by Babones (2008), Fuse and Crenshaw (2006), Jorgenson and Rice (2010), Moore et al. (2006), O'Hare et al. (2013), Pamuk et al. (2011), Shandra et al. (2004, 2005, 2011), Shen and Williamson (1997, 2001), and Wang (2014) have on many occasions show that as countries grow to become modernized, the rate of death of new born babies reduce. However, some other scholars, such as Amand and Ravallion (1993) and Frey and Field (2000), challenge this stand, opining that the connection is not always straightforward.

2.2.4 Human capital theory

Schultz (1961) and Becker (1964) emphasized that improvements in productivity and long-term economic growth are strongly linked to investments in human capital—particularly education, skills, and health. The theory promotes the idea that individuals are strategic when deciding to improve their well-being and future earnings, often by investing in both physical and human capital. potential investments are expected to yield returns in the form of higher income, better working conditions, good health, and longer life expectancy. O'Mahony and Samek (2016) noted that a healthy workforce sustains labor supply and productivity, as healthier workers are better positioned to acquire new skills and extend their working lives. on the other hand,

morbidity reduces productivity through absenteeism and early retirement. More recently, empirical evidence highlights the role of women's empowerment in reducing infant mortality in Nigeria, presenting the need to encourage female labor participation through better education and healthcare (Zheng *et al.*, 2024). Macroeconomically, human capital is recognized as a critical factor of production, where health and education jointly determine productivity and economic performance (Lim *et al.*, 2018). Based on ideology of Becker (1964) that well-being is an unavoidable necessity and an essential aspect of human capital that must be ensured. Infant mortality stands as a future decline in the human capital resulting from poor macroeconomic state, because a higher rate of this phenomenon reduces the number of individuals who are available to participate in future economic activity. In the inquiry, the income levels, rate of electricity access, public health financing, and foreign supports which are the reflection of the macroeconomic status in the country are reviewed as they may stimulate a stronger human capital accumulation and lasting economic transformation as a result of productive input. Despite criticisms that the theory is overly simplistic, governments continue to prioritize investments in health and education as strategic tools for growth. Infant mortality in this case would witness a lasting decline if the authority channels the right resources with the required supervision in the health sector to ensure that the goal is achieved without disruption.

2.2.5 Health Led Growth Hypothesis (HLGH)

The health-Led Growth Hypothesis, introduced by Mushkin (1962), view health as an important agent of economic growth, part of endogenous growth theory. It argues that investing in health is not only a social benefit but also a key economic strategy, as healthier populations enhance human capital. Improved health boosts productivity, extends working life, and reduces

1 absenteeism, leading to sustained economic performance. Health spending is seen as capital
2 investment, yielding direct returns in labor productivity and indirect benefits like improved
3 quality of life and reduced infant mortality. This hypothesis projects health as both a driver and
4 a result of economic growth, stimulating a cycle of enhanced population well-being and national
5 development. The hypothesis postulates that investment aimed at the development of the health
6 sector and improvements in healthcare delivery tend to positively impact economic growth. The
7 idea behind this hypothesis is that the healthier a national population the higher the human
8 capital and productivity. This creates a promising potential for increased output, increased labor
9 productivity, and demographic dividends (Klotzbach, *et al.*, 2003). The sub-Saharan region of
10 Africa for instance faces negative economic possibilities as a result of the disease burden.
11 Erdogan and Erdogan (2024) disclosed that the model is worthy to be considered in the
12 formulation of policies that target health and development in the west African area of Africa.
13 This is very necessary as it would aid their plans to attain far-reaching global development
14 plans. The investigation revealed that the influence health expenditure has on economic growth
15 is bidirectional. This empirical evidence translates to the assumption of the hypothesis because
16 good state of health determines the productivity of labor by reducing absenteeism and
17 enhancing individuals' performance. However, health investments can improve life expectancy,
18 incentivize long-term economic planning, and reduce external dependency, thereby supporting
19 economic growth. Similarly, Artekin and Konya (2020), in a health-led growth hypothesis
20 (HLGH) based study confirmed that investments in health sectors of selected OECD countries
21 through deliberate health expenditures. In the long run, the outcome showed a substantial
22 economic benefit as a result of improved overall quality of human capital. The Health Led
23 Growth Hypothesis (HLGH) is measure suggested for policy makers to make health investment

1 a top priority in achieving sustainable economic growth. Thus, The Health Led Growth
2 Hypothesis would provide a reliable framework in understanding the influence health
3 macroeconomic variables has on Infant Mortality Rate.

5 **2.3 Empirical Review**

6 In the quest to discover the many pitfalls in the realm of healthcare and to proffer the best
7 possible solutions, numerous researchers in their works show how macroeconomic
8 determinants such as –Gross Domestic Product Per Capita (GDPPC), Government Healthcare
9 Expenditure (GHE), Inflation Rate (INF), and Access to Electricity (ACE) as they relate to state
10 of health indices such as - life expectancy, access to healthcare, and infant mortality rate etc. As
11 it is highlighted below, these previous studies have critically laid foundation and hence, this
12 inquiry aims to discover the stones left unturned in this field.

13
14 Miftahu and Kawu (2024) screened healthcare investment and how it influences infant mortality
15 in Nigeria using the Autoregressive Distributed Lag (ARDL) and Bound Testing methods. The
16 inquiry laid its focal point on the long-run and the interaction between public healthcare
17 expenditure and new born deaths, depending on child support programs, immunization,
18 nutrition, and education. The outcome proved a significant long-run connection, showing that
19 increased healthcare investment leads to fall in new born deaths in Nigeria. The authors relied
20 on this outcome to prescribe that the government should strengthen efforts to mobilize and
21 properly monitor the disbursement of funds for child-focused health programs in order to
22 promote child rights, improve child development, reduce infant mortality, and enhance overall
23 well-being

1 Alawiye (2024), researched on how telemedicine, electronic health records, and mobile
2 applications make the delivery of health care easy. This study focused on ascertaining the ways
3 these life-made-easy tools drive efficiency, improve patient health, and to also identify the
4 problems user face while trying to master the new tech. Following a rigorous screening, the
5 result showed a significant digital technological influence on patient due to easy-to-use
6 prompts, efficiency, and accuracy. Furthermore, the presence of digital technology brought
7 about protected patients' personal information, connectedness of the systems, and low cost of
8 installation thereby reducing persistent challenges in the Healthcare system and improving
9 infant mortality.

10

11 Li and Li (2024), in a study connected Government health expenditures and neonatal mortality
12 on how the former influence the latter. The idea was to trace the progress of meeting up with
13 the sustainable development growth 3 (SDG 3) using China as a case study. This investigation
14 was aided using comprehensive graphical, descriptive and econometric methods. the result
15 showed evidently that China has successfully met SDG 3 as a result of suspected increased
16 government health expenditure.

17

18 Guo *et al.* (2024) made use of panel ARDL and quantile regressions to understand how CO₂
19 emissions, energy use, and urbanization affect public health. The outcome of this observation
20 showed that renewable energy, urbanization, GDP, and industrialization reduce infant mortality,
21 while CO₂ emissions increase both in fewer years and far-reaching periods. The prescription
22 suggests a sustainable urbanization to improve access to health facilities and strengthen public
23 health.

1 Mishiwo *et al.* (2023) investigated how infant mortality shocks affect economic growth in the
2 Nigerian State using a multiple linear regression model. The outcome of this investigation
3 disclosed a significant negative relationship, showing that higher economic growth leads to a
4 fall in the deaths of new born babies.

5

6 Udemba *et al.* (2023) carried out a screening on energy sources and subsidies and how they
7 interact to influence new born demise in India using Residual Augmented Least Squares and
8 Engle and Granger techniques. The observation exposed that renewable energy, higher income
9 per individual, and public subsidies caused a dwindle in infant mortality, while fossil fuels, CO₂
10 emissions, and private subsidies increase it.

11

12 Awoyemi *et al.* (2023), investigated how government expenditure affects health outcomes in
13 Nigeria for a period of 23 years (1995-2018). Using autoregressive Distribution Lag approach.
14 The inquiry resulted to a case of negative connection among the core variables, recommending
15 higher public funding and strengthening public-private partnership to improve the health care
16 delivery.

17

18 Koginam and Anthony (2023), Used Autoregressive Distributed Lag approach to study how
19 different mortality indices affected Real GDP in a long-run. Outcome of the screening disclosed
20 that children between the age of 1 and 5, adult male and women mortality, new born deaths all
21 negatively and significantly influenced RGDP in the model in longer periods. However, in the
22 shorter period, new born deaths had negative but statistically significant influence on RGDP.

23

1 Miladinov (2023), examined how the combination of social and economic factors causes infant
2 mortality in Macedonia, Turkey and Albania from 1991-2017. The investigation revealed that
3 the reduction in infant mortality rates across all three countries and these factors share a strong
4 negative a strong negative connection. Using fixed-effect and the system of equations along
5 with seemingly unrelated regression method, the result confirmed the significance of the
6 variables used.

7

8 Jiang and Liu (2022) looked at the causal link between general price level and the deaths of
9 new born babies in China employing Mixed-Frequency VAR (MF-VAR) model supported with
10 impulse response and variance decomposition analysis, and compared it with a low-frequency
11 VAR (LF-VAR). The findings showed a clear causal relationship, with the Consumer Price
12 Index (CPI) exerting a stronger influence on new born death rates while prescribing price
13 stability as a one of the importance factors necessary for achieving SDGs.

14

15 Asghar *et al.* (2022) examined how the provision of certain amenities like electricity in rural
16 and urban areas coupled with the consumption of biomass energy and how they connect to the
17 deaths of infant mortality. The observation covered the period between 1990 and 2020 using
18 the Panel Quantile Regression (PQR) method and the outcome show that biomass energy
19 consumption increases infant mortality, while electrification shows an inverted U-shaped
20 relationship across different quantiles. the research advised reducing reliance on biomass and
21 promoting affordable, accessible electricity to lower new born death rates.

22

1 Bao *et al.* (2022) scrutinized how of housing rentages and the rising general price levels affect
2 the people living in the following countries: Canada, France, Japan, Netherlands, Spain,
3 Switzerland, Sweden, UK, and USA For a 23 years period (1995-2019). The observation
4 utilized two-stage least squares (2SLS) and the generalized method of moments (GMM) to
5 address the situation, and the outcome showed that rentages share positive and significant
6 influence on infant deaths while extends time one is expected to live. Inflation was shown to
7 increase occurrence of infant death and reduce life expectancy, whereas GDP and health
8 expenditure contributed to better health outcomes by lowering infant mortality and raising life
9 expectancy. However, unemployment exhibited a harmful effect on population health and the
10 research recommends that Health practitioners consider the housing market and role of general
11 price levels.

12
13 Adeosun *et al.* (2022) carried out an inquiry into how certain economic tools like: income per
14 capita, inflation, and population size on Nigeria's mortality rate. The study confirmed
15 stationarity at orders $I(0)$ and $I(1)$ evidence of co-integration among the tested variables. The
16 investigation showed a finding where in the short-run the connection among income per capita
17 and death rate proved positive. This indicating that rising income does not immediately cause
18 a fall in mortality rate. Similarly, inflation positively affected mortality in the short periods,
19 while no long-run relationships were established.

20
21 Aniefor (2022) investigated the influence of company income tax on death rate of new born
22 applying descriptive analysis and regression. The result of this investigation proved negative

1 with an insignificant effect, and recommended revising tax incentives to attract investment,
2 boost revenue, and fund better healthcare for mothers and children.

3
4 Hoque and Rana (2022) studied how social factors and certain tools that measure economic
5 performance interact with infant death rates in Bangladesh and India using a Vector Error
6 Correction Model Approach. Their findings showed that GDP, funding education, and role of
7 women in production influenced infant death rates negatively. Whereas the unemployment rate
8 and female school enrollment were revealed to improve infant mortality rates.

9
10 Logarajan *et al.* (2022) screened how sufficient funding of health care influence mortality of
11 between new born babies and children aged 5 years in Malaysia using ARDL technique. The
12 result showed that private spending on health increases rate of under-five mortality whereas the
13 public and private spending are statistically insignificant suggesting an effective health
14 financing safety net to encourage an improved child health outcome.

15
16 Akpesiri and Eburajolo (2022) carried out a screening to understand how monetary policy
17 connects to infant death rate in Nigeria, covering the period 1980 to 2016. The investigation
18 applied Vector Error Correction Model (VECM). Outcome from the screening disclosed that
19 literacy rate and per capita income caused the fall of infant mortality though not significantly,
20 and the study prescribed a stronger monetary policy to stimulate educational investment and
21 sustained economic growth for improved health outcomes.

1 Mohammed and Akuoko (2022) investigated how access to electricity at the subnational level
2 affect infant mortality rate in Ghana. The research utilized pooled cross-section regression
3 approach and discovered fewer incidents of infant mortality in regions with an improved access
4 to electricity while on the contrary, the opposite is the case in the regions with epileptic power
5 supply. The study recommends reliable access to reliable electricity coupled with health
6 infrastructure development policies to improve healthcare issues.

7
8 Adeleke *et al.* (2022) employed spatial statistic as the technique to examine how
9 macroeconomic factors affect under-5 mortality in Nigeria. It was discovered that in the
10 northern region of Nigeria, some areas exhibited high rates of under-5 mortality which is linked
11 to variations in the internally generated revenue where as in the southern region of Nigeria, the
12 inflation rate is observed to influence the under-5 mortality rate negatively.

13
14 Doucouliagos *et al.* (2021) in an effort to find out the effect health aid has on infant mortality
15 analyzed panel data from 96 countries from the period 2002–2015. Outcome of this
16 investigation showed that health aid can significantly reduce infant deaths, while
17 recommending that its effectiveness depends on the quality of governance, noting that recipient
18 countries must strengthen governance to maximize impact.

19
20 Bugelli *et al.* (2021) analyzed regional inequality and factors that affect infant mortality in
21 Brazil across 26 states for the period (2004–2015). The investigation in that vein applied
22 multilevel panel data coupled with fixed effects to screen the influence of employment rate,
23 household income, and health capabilities. Findings showed that, apart from known factors like

1 the Bolsa Família Program and fertility rates, educational attainment, quality of prenatal care,
2 and access to health professionals significantly affected infant mortality. however, employment
3 rate was linked to neonatal mortality up to three years, while families earning below two
4 minimum wages faced a higher risk of infant deaths.

5
6 Rahman and Alam (2021) analyzed data from 2002–2018 across SAARC-ASEAN countries to
7 understand the connection that exist between electricity access, women’s literacy, and spending
8 with regards to women well-being. The study used advanced panel techniques and discovered
9 that electricity access, female education, health spending, economic growth, and immunization
10 rates all improved female life expectancy and reduced female adult mortality. A one-way causal
11 relationship was also established among the variables.

12
13 Adeosun and Faboya (2020) carried out unique research on what healthcare spending and child
14 mortality relationships may look like in Nigeria. Outcome of this screening led to a negative
15 but insignificant connection among the indicators tested which simply translates to spending on
16 healthcare to a small degree play a part in the reduction of the death rate of new born child in
17 Nigeria.

18
19 Kara and Ersin (2020) inquired to find out the potential connection spending on health by the
20 authorities could have on infant death rates in 20 OECD countries between 1980 and 2017 by
21 utilizing Kao panel cointegration and Dumitrescu-Hurlin causality tests. What was discovered
22 in the investigation was that in the long-term model, the connection among the variables proved
23 positive.

1 Karaman *et al.* (2020) investigated OECD countries using stepwise multiple regression to
2 assess how healthcare spending impact health outcomes. What was discovered from the inquiry
3 showed a significant impact by health care spending by the authorities per person on affects
4 nursing mothers' deaths, new babies' deaths, and as well as time one is expected to live. whereas
5 private spending showed a strong influence on self-reported health. The authors recommend
6 increasing investment on matters concerning mother and child health to reduce the tragedy.

7

8 Boachie *et al.* (2020) in a study observed the importance of augmenting health funding by the
9 government in countries that are still developing. fixed effects technique was employed for
10 analysis during the inquiry from 1995-2014. Outcome of the observation revealed that
11 government spending on health-related public goods strongly lead to a fall of infant death rates.
12 It was further showed that improvements of social and economic amenities like drinkable water,
13 income per capita etc. results to reduced health outcomes such as infant deaths reduce.

14

15 Guo *et al.* (2020) preened economic uncertainty and the role it plays in shaping humans state
16 of health in China. The observation took the initiative by applying Quantile Auto Regressive
17 Distributive Lag (QARDL)model to execute the analyses. The result showed both models to
18 exhibit short and long connections, where new born death rate exhibited positive and significant
19 at all quantiles. However, negative shocks of economic uncertainty exerted reasonable degree
20 of opposite directional influence on new born death rate at the highest quantiles.

21

22 Raji (2020) used the Fully Modified OLS model to investigate how macroeconomic factors
23 influence the degree of infant deaths and children under the age of 5 from the 1986 to 2017 in

1 Nigeria. Outcome of this observation revealed that unstable macroeconomic policy led the
2 following indicators: inflation rate, unemployment, and exchange rate volatility to greatly
3 increase the death rates of new born babies and children below the age of 5.

4
5 Ebeh *et al.* (2019) assessed the impact of infant mortality on economic growth in Nigeria using
6 the ARDL model. The result of the study revealed a significant negative effect of infant
7 mortality on growth in both the short and long run. the recommendation suggests that a stronger
8 government–private partnerships is key to achieving sustainable growth and to reduce infant
9 mortality in line with the SDGs.

10
11 Shobande (2019) examined 23 African countries (1999–2014) using the Becker hypothesis and
12 Grossman models. in the end of the screening, the outcome came out and the measured
13 predictors showed a significant negative connection with the death rate of new born babies.

14
15 Akinlo and Sulola (2019) assessed 10 nations in the lower part of African Saharan from year
16 2000 to 2008. Outcome of the assessment highlighted that the following indicators: income per
17 head, health aid, degree of HIV Transmission, led to the deaths of new babies in a significant
18 negative manner.

19
20 Ubi and Ndem (2019) employed the Vector Autoregressive Econometric (VEC) approach to
21 study whether health outcomes in Nigeria, specifically life expectancy and infant mortality, are
22 affected by poverty shocks. The findings revealed that poverty shocks do not directly affect
23 these health outcomes. Instead, the variations are largely caused by shocks within the health

1 system itself. The study advised that improving health outcomes is key to achieving poverty
2 reduction.

3

4 Attamah (2018) applied the VECM mechanism to analyze the inquiry of whether government
5 health funding connected to the persistent demise of new born babies in Nigeria and the inquiry
6 spanned from 1981 to 2018. Outcome of this inquiry aligned with what was expected as the
7 connection proved a significant positive. Suggestion prescribed is the implementation of a
8 sustainable health scheme funded jointly by the federal, state, and local governments to improve
9 child health outcomes.

10

11 Heshmati (2018) examined the convergence of OECD countries in terms of GDP and healthcare
12 expenditure per capita using an Augmented Solow Model that incorporated health capital. The
13 analysis revealed that health expenditure and investments positively contribute to economic
14 growth, with a close link between GDP and health spending. The regression results proved the
15 interconnectedness of health expenditures and healthcare system performance across countries.

16

17 Hashiani *et al.* (2015) examined panel data from 196 countries to understand how economic
18 inequality impact infant mortality, using indicators such as income per person, GNI, privately
19 spent monies, and government funding as per person. Outcome of the examination disclosed
20 that new born babies, children under age five, and nursing mothers' death rate were highest in
21 undeveloped countries. Among the economic indicators, government funding as per person,
22 income per person, and GNI were shown to be crucial in explaining disparities in mortality,
23 while the out-of-pocket index had no significant effect.

2.4. Gaps in the Literature

Idea that propelled this inquiry of connecting income levels, rate of electricity access, public health financing, and foreign supports to the persistent death of new born babies in Nigeria will particularly surround on technological advancement, measured through access to electricity, as a driver of effective healthcare delivery and infant survival. While previous studies have majorly focused on the indirect impacts of electricity on health outcomes such as through economic growth, education, or living standards there is limited empirical research directly link electricity access to infant mortality. Also, little attention has been paid to how government policies and programs designed to improve electricity access are implemented and how they subsequently influence health outcomes. In addition, the potential role of renewable energy solutions in reducing infant mortality in underserved or off-grid areas remains ignored.

This research further distinguishes itself by incorporating four macroeconomic variables—National Income per Capita (GDPPC), Government Health Spending (GHE), Electricity Access (ACE), and Health-Specific Aid. These variables have been rarely examined together, as earlier studies typically considered only one or two, often excluding technological factors such as electricity access. Also, this inquiry will provide the latest analysis, based on the period from 1990 to 2024, thereby appending the scope of this study to the most recent related study by Awoyemi *et al.* (2023), which covered only 1993–2018. By addressing these gaps, the inquiry makes a unique addition to the literature and is expected to generate findings with strong policy relevance for Nigeria’s health sector.

CHAPTER THREE

METHODOLOGY

3.1 Research design

McCombes (2021) explains that a well-structured research design is important because it ensures that the style of technique used by someone for research purposes correspond goal, they intend to unravel such that the analysis applied is suitable for the data. Research design, therefore, provides a blueprint that guides how data is gathered, organized, and analyzed in order to achieve reliable results (Ogwuche, 2021). In other words, it simply means the strategy a researcher uses to accomplish the objectives of a study, while also reflecting the important decisions made before and during data collection.

The design applied in the inquiry was retroactive and quantitative in nature due to the ex post facto rule. Subsequently, econometric modeling was applied to carry out the analysis with the intention to discover the nature of connection these chosen macroeconomic variables have with infant mortality rates over long period of time. The ex-post facto approach is particularly relevant because it investigates events that have already occurred, allowing the researcher to examine naturally existing conditions without manipulating the explaining parameters. By understanding the mechanism, the inquiry was able to explore the link among deaths of new born babies and the chosen variables namely: Income Per Capita (GDPPC), Government Healthcare Expenditure (CHE), Access to Electricity (ACE), and Health-Specific Aid (AID). This design therefore provided a reliable basis for observing long-term trends and understanding how these macroeconomic factors have in the past shaped health outcomes in Nigeria.

3.2 Theoretical Framework

This inquiry applied the complex mechanism of how an economy grow exogenously as crafted by Solow and Swan (1956). The neoclassical design fundamentally states that for an economy to grow, three factors are to be considered such as – labor, capital, and technology. Additionally, this growth theory emphasized the importance of accumulation of capital in an economy and how this capital is utilized due to the issue of scarcity. The efficient use encourages economic growth and hence, determines the performance and productivity of labor. Technology is a crucial factor in neoclassical growth theory, since the more it is acquired and incorporated in the process of production, economic growth is assured. Therefore, it is necessary that government intervenes via investing in scientific and research development towards innovation (Drago lava *et al.*, 2016).

This framework will center on Solow-Swan growth theory from which this study will draw its hypothesis that improvements in electricity access contribute to better infant mortality rate by enhancing productivity of health services, improving living conditions, and economic flourish. The blend of the parameters as used in this investigation will rely on the craft by Solow and Swan (1956), to give this research smooth sailing and effectively discover the underlying connection among the chosen macro-economic variables and the infant mortality in Nigeria. The key concepts: Capital Accumulation –investment in the physical capital such as healthcare infrastructures, equipment and facilities, grows the capital stock which in return leads to higher output levels; Labor growth – the increment in the labor force leads to economic output growth; Technological Progress – upgrades in extant technology and acquiring new technologies enhances productivity and economic growth; Diminishing Returns – returns on capital and labor

diminish as extra of each is added settling to constant and equilibrium state where output growth equals the technological progress. This inquiry relied on Corporate Finance Institute (2024), by applying the functional form of production, as mathematically crafted by Charles Cobb and Paul Douglas. Thus, the following equation:

$$Y = A \cdot K^{\alpha} \cdot (L)^{1-\alpha}$$

3.1

Where: Y denotes the GDP, A stands for the determinants(s) of technology, K denotes the capital stock, L stands for labor inputs, and Alpha (α) represents the output elasticity of capital usually standing between zero and one.

The inquiry aimed to dissect the connection of selected macroeconomic variables and how they may decide demise of infants, placing particular emphasis on how health integrates into human capital development. Health Outcomes such as infant mortality can be influenced by capital accumulation for example government health expenditure and building of infrastructures, labor growth for instance the health status of the populace, and Technological advancement such as – steady electricity and medical innovations (Makiew *et al.*,1992). To show how infant mortality fits into the inquiry, it was adapted to the traditional Solow growth model which was transformed to accommodate the human factor along with labor. Infant mortality is considered a determinant of human capital, for instance, increased infant demise is equal to future reduction in the labor force, productivity, and a lasting economic growth and vice versa. Hence, the modified model for infant mortality rate:

$$Y = A \cdot k^{\alpha} \cdot (H \cdot L)^{1-\alpha}$$

3.2

Where: Y denotes for Health outcomes (infant mortality rate), A captures the effect technological advancement, k denotes capital (government health expenditure and infrastructures), H denotes human capital (population health status), and L denotes labor (population size), and $\text{Alpha}(\alpha)$ stands for output elasticity of capital.

3.3 Nature and Sources of Data

This inquiry, in the quest of discovering how these chosen indicators decide the rate of infant demise, expanded its radar and search scope to a 33-year period from 1990 to 2024. The period spanned would provide a critical understanding of how key macroeconomic indicators influence the demise of new babies in Nigeria. Data for the independent variables - Gross Domestic Product Per Capita (GDPPC), Health Spending (CHE), Electricity Access (ACE), and Health-Specific Aid (AID) as well as the dependent variable – Infant Mortality Rate were gotten from macrotrends, a subsidiary of the World Bank. This is a credible institution and hence, ensures reliability and validity of the dataset while giving the study a robust finding.

3.4 Model Specification

To determine the short-run or how far-stretched the influence of the choice indicators on new born babies' death rate occur in Nigeria, the inquiry applied Autoregressive Distributed Lag (ARDL) model. The ARDL technique is especially viable in testing for cointegration in small samples and is easy to use in its application. The capacity of this technique will give the inquiry room to discover if the parameters are integrated at order $I(0)$, $I(1)$, and also show if both existed (Ogwuche, 2021). Also, the inquiry will make mathematical specification based on the theoretical ideas as crafted by Solow and Swan (1956), which emphasizes the role of macroeconomic factors—particularly technological progress—in shaping output and overall

1 economic growth. Based on this framework, the study seeks to empirically examine how key
 2 macroeconomic variables influence infant mortality in Nigeria. Accordingly, the mathematical
 3 expression that shows the study's hypotheses and expectations is specified under the ARDL
 4 framework as follows:

$$5 \quad IMR = f(SMV) \quad 3.3$$

6 Where:

7 IMR = Infant Mortality Rate

8 SMV = Selected Macroeconomic Variables

9 Factoring in the variables for Health Macroeconomic determinants, gives:

$$10 \quad IMR = f(GDPPC, CHE, ACE, AID) \quad 3.4$$

11 Where:

12 $GDPPC$ = GDP Per Head

13 CHE = Capital Spending on Health

14 ACE = Access to Electricity

15 (AID) = Health-Specific Aid

16 Setting up equations in econometric forms gives:

$$17 \quad IMR = \beta_0 + \beta_1 GDPPC + \beta_2 CHE + \beta_3 ACE + \beta_4 GHE + \mu_1 \dots\dots\dots 3.5$$

18 Where: β_0 stands for the intercept or the constant value or the autonomous variable estimates
 19 for the Infant Mortality Rate; $\beta_1 - \beta_4$ are respective estimates for the selected namely: GDP
 20 Per Head ($GDPPC$), Capital Government Health Expenditure (CHE), Access to Electricity
 21 (ACE) and Health-Specific Aid (AID), and μ_1 stands for the stochastic value.

22

23

3.4.1 A priori Expectations

Ogwuche (2021) defined this process as an attempt made to connect the explained parameter with the independents used in an inquiry as envisaged by the theory that will be used as framework. It serves as guide for a researcher as they juxtapose their empirical finding to their a priori expectation to examine the validity of their findings and for clear interpretation. Hence, the parameters are examined to make sure it aligns with the stand of the theory. In this study, the a priori expectation shed light on the anticipated outcome or the predicted outcome in line with the existing theory and knowledge how death of infant death rates and macroeconomic tools selected in the inquiry such as - GDP Per Head (GDPPC), Capital Government Health Expenditure (CHE), Inflation Rate (INF), Access to Electricity (ACE), and Health-Specific Aid (AID) which are the independent variables.

Table 3.1: Summary of the Expectations

Variable	Acronym	Connections with IMR
GDP Per Head	GDPPC	-
Capital Gov. Health Expenditure	GHE	-
Access to Electricity	ACE	-
Health-Specific Aid	AID	-

Source: *Author's Compilation, 2025.*

From the above *a priori* table, a higher income per capita (GDPPC) provides households the capacity to afford nutritious food, access to good education, and access quality healthcare resulting to a reduced IMR. The connection here with new born demise and income per head is a negative, hence, $\frac{\Delta IMR}{\Delta GDPPC} < 0$. If the government spending on health increases, it is expected that child health care and would improve therefore reducing the IMR. But on the other hand, when supervision is ignored, corruption and mismanagement would leave health sector in disarray. The expected relationship is negative, thus, $\frac{\Delta IMR}{\Delta CHE} < 0$. Reliable access to electricity

1 brings about efficiency in healthcare and services such as improved infant care and better
 2 storage of vaccines thereby reducing IMR. But on the contrary, an epileptic national grid will
 3 yield a negative result. The relationship as the theory postulated would appear to be negative,
 4 this is symbolized as $\frac{\Delta IMR}{\Delta ACE} < 0$. The expected relationship between health-specific aid and infant
 5 death is that as aid increases, infant mortality rate drops and vice versa. Thus, this is symbolized
 6 as $\frac{\Delta IMR}{\Delta AID} < 0$.

7

8 **3.5 Variable description and measurement**

9 At this juncture, Table 3.1 displays the parameters and how it is calculated, this is shown below:

10 **Table 3.2: Variable Description and Measurement**

Variable	Acronym	Measurement	Description
Infant Mortality Rate	IMR	Deaths per 1000 birth	The infant deaths that occurred within after 365 days of 1000 new births.
Gross Domestic Product Per Capita	GDPPC	US Dollars in constant prices	The total of the national income spread across the population in other word, average income per person.
Capital Government Health Expenditure	CHE	Percentage	Total funds spent on health as the percentage of GDP.
Access to Electricity	ACE	Percentage	The proportional distribution of electricity to the population.
Health-Specific Aid	AID	US Dollars	The proportion of total aid received by the health sector.

11 **Source:** *Author's compilation, 2025*

12

13 **3.6 Techniques for Data Analysis**

14 The following procedure were used to understand how the selected macroeconomic variables
 15 connect to new born babies' death rate in Nigeria (1990-2024). These techniques are used to

answer the investigative questions; actualize objectives; and establish how these macroeconomic tools connect to the deaths of new born babies in Nigeria.

3.6.1 Variables trend analysis

The variable trend analysis was conducted as it is important in identifying and understanding the trends and patterns of the variable for the observed period. This technique is useful because it provides a comprehensive pattern for each of the selected macroeconomic variables (GDP Per Head, Capital Spending for Health, Access to Electricity, and Health-Specific Aid) and how each connect to the demise of new born babies in Nigeria. This was possible as a historical data on these variables were obtained to project a graphical analysis of the trends and patterns of the observed period of (1990–2024). Notwithstanding, this study did not overlook the roles of electricity, external factors such as SDGs, National Policies such as - National Health Policy; Child Health Policy; National planning; Child health program guidelines strategies and how it influences Infant Mortality Rates (IMR).

3.6.2 Descriptive statistics analysis

According to Ataboh *et al.* (2023), descriptive statistics analysis gives a comprehensive approach to summarization, organization, and presentation of a dataset. Hence, this study on empirical analysis of the selected macroeconomic variables on infant mortality rate (1990-2024) used descriptive analysis to summarize important characteristics of the variables. The

following statistical tests were carried out: to measure central tendency mean: $\bar{X} = \frac{\sum X_i}{N}$ where:

X_i stands for a number, N is the total number used, and mode were applied. For dispersion also

known as spread of data, the study utilized standard deviation, it is expressed as: $\sigma = \sqrt{\frac{\sum (X_i - \bar{X})^2}{N-1}}$

where “ X_i ” stands for each x value, “ $\bar{X}$ ” is the value for Average, and “N” stands for time covered. Variance and range were also carried out. Furthermore, normality was calculated during this inquiry alongside skewness and kurtosis measures. Skewness was used to evaluate the degree of asymmetry in the data distribution, while kurtosis provided insight into the peakedness of the distribution relative to a normal curve.

3.6.3 Correlation analysis

To understand how the pattern of the connection looks like taking into consideration degree and direction, the inquiry employed Correlation Analysis. The nature of the examination was bivariate which enabled clear understanding of the connection degree among core indicators focused on in the inquiry, and the direction of this relationship was established given the values of the correlation coefficients. The coefficients are expected to range between +1 and -1. A value of +1 implies that the magnitude of the association among the variables is perfect degree and spells a positive correlation whereas on the contrary, a value of -1 shows a weaker correlation and means a negative correlation, but if the value gotten is 0 it means there is no correlation. This analysis is mostly measured using the central formula stated as: $r =$

$$\frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2} \sqrt{\sum(Y_i - \bar{Y})^2}} \quad \text{where: } r \text{ is expected to range from } -1 \text{ to } 1, X_j \text{ and } Y_j \text{ are the respective}$$

infinite values of the data, $\bar{X}$ and $\bar{Y}$ are the averages of the two actual data, while the numerator represents the covariance among X and Y, whereas the denominator normalizes the values using their standard deviations.

3.6.4 Unit root test

In this inquiry, the data collected spanned the period 1990 to 2024. Usually, the issue with this form of data is the likelihood of unit root, that is to say that it exhibits non-stationarity and a wandering behavior or random walk. It was essential to perform a difference stationary process to identify this issue and duly correct the unit root to ensure that the data collected for new born demise rate in the collected period along with the chosen macro-economic tools in the specified regression model is stationary and to ascertain a valid result. As Traore (2022) noted, the presence of unit root in an econometric modelling may have a serious implication and defeats the purpose of a regression analysis as the result can become spurious, inflated R-Squared values, and abnormally low Durbin-Watson statistic. The inquiry also applied the ADF test crafted by Dickey and Fuller (1981) to address this issue. The general form of the craft is written as follows: $\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{i=1}^j \delta_i \Delta Y_{t-1} + \varepsilon_t$ where: Y represents infant demise rate per this inquiry, ΔY_t stands for the value at first difference of Y_t , t stands for the time, $\sum_{i=1}^j \delta_i \Delta Y_{t-1}$ is for the lagged differences as toggled in the test to avoid autocorrelation, while epsilon stands for the disturbance. The value gotten from the computation with stationarity at 5% level of significance would qualify for use in this inquiry. Maintaining stational time series is crucial, to avoid the incident of spurious results and ensure meaningful econometric modeling in this study.

3.6.5 Cointegration Test

Given the aim of this study, one of the objectives was to screen into a far-reaching period to enable clear understanding of how new born demise and the following indicators – GDP Per Head (GDPPC), Government Health Expenditure (CHE), Access to Electricity (ACE), and

Health-Specific Aid (AID). Cointegration testing approach as crafted by Pesaran *et al.* (2001) was applied in this inquiry. During the procedure, the study assumed a null hypothesis of no cointegration opposing the alternative hypothesis of cointegration among the variables. The process of this computation was that at some point, one would choose a value based on the comparison with F-statistics and tabulated critical bounds values at levels $I(0)$ and $I(1)$ and this depended mainly on the available level of significance, an outcome of a no co-integration would qualify as a far-reaching connection and vice versa. Moreover, the computation followed two-step procedure as crafted by Engle and Granger (1987), which at the end solidified its outcome. The general ADF form applied to the residuals is given as: $\Delta\varepsilon_t = \gamma\varepsilon_{t-1} + \sum_{i=1}^j \delta_i \Delta\varepsilon_{t-i} + \varepsilon_t$ where: $\Delta\varepsilon_t$ stands for the first difference of residuals, γ is the coefficient used to test stationarity, j stands for the lag length, ε_t is the error term.

3.6.6 Autoregressive distributed lag (ARDL) technique

The main idea that's driving this inquiry was to screen deep into the connection that may exist between infant demise rate the selected macro-economic indicators, to discover how far it stretches and this was aided by the framework provided by ARDL. According to Kripfganz and Schneider (2018), ARDL models are useful when analyzing the dynamic relationship of a time series because it establishes a comprehensive short-run and long-run relationship among the variables while accommodating their mixed stationarity. The mathematical expression for this Technique for establishing the impending connection of the chosen macroeconomic variables with infant mortality can be expressed as:

$$IMR_t = \beta_0 + \sum_{i=1}^p \alpha_i IMR_{t-i} + \sum_{j=0}^q \beta_j X_{t-j} + \mu_t \quad 3.6$$

Where: IMR_t = Dependent variable (Infant Mortality Rate current), β_0 = Intercept value

$\beta_j X_{t-j}$ = coefficient of lagged independent variables (GDP Per Head (GDPPC), Gov. Capital spending (CHE), Access to Electricity (ACE), and Health-Specific Aid (AID) lagged by (j) periods, α_i = Coefficient of the lagged IMR or it's past effect, μ_t = Error term, and p & q = the number of lags.

3.6.7 Error correction model (ECM)

Following the outcome of applying ARDL method for estimation to understand the nature of the connection relative to the objectives of the inquiry, it is important to highlight another component of this technique. The component ECM records how fast an economy recalibrates back to stability after some disturbance which can come in the form of an economic shock or short-run disequilibrium. According to Pesaran and Pesaran (1997), Pesaran and Smith (1999), and Pesaran *et al.* (2001), the ARDL techniques involves two stages. The first thing that happens when ARDL is carried out is that the executes bounds testing to confirm if the macro-economic tools relate to the deaths in in longer periods. While in later stage, it combines estimation for shorter period elasticities and the long-run coefficients, which comes with error correction value. Generally, the ECM is computed using the following:

$$\Delta Y_t = \beta \Delta X_{t-1} + \beta EC_{t-1} + \mu_t \quad 3.7$$

Where: $\beta \Delta X_{t-1}$ = the lag (period one) of the variables and βEC_{t-1} = The error correction value.

In the case where the long-run relationship and association among the variable has been established, then it's necessary that this study commences with the assessment of the long-run effect and the short-run state using the Error Correction Model approach to figure out the speed

of adjustment. Thus, in this analysis of the impact of selected macroeconomic variables on infant mortality in Nigeria, the ECM is expressed as:

$$\Delta IMR_t = \beta_0 + \beta_1 IMR_{t-1} + \beta_2 \Delta GDPPC_{t-1} + \beta_3 \Delta CHE_{t-1} + \beta_3 \Delta ACE_{t-1} + \beta_4 \Delta AID_{t-1} + \sigma EC_{t-1} + \mu_t \quad 3.6$$

Where: Δ = First difference operator, β_0 = Constant parameter, β_1 to β_4 = Parameter Co-efficient, and EC_{t-1} = degree of shock in an economy with recalibration speed

3.6.8 Estimation medium

This study utilized Econometric View 12th edition (EViews 12) to carry out the necessary estimations and diagnostic tests. The reason for the adoption of this statistical software was that it facilitated the complex tests conducted by the author electronically.

3.7 Post-estimation Tests

3.7.1 Normality test

Normality test is essential for the measurement of errors in this scientific experiment. it is a probability distribution that is attributed by a bell -shape and symmetrical around its mean with the outcome expected to exhibit respective values of 0 and 3 (Jarque & Bera, 1980). This study made use of Jarque-Bera test to ensure that at the end of this computations the results come out accurate per the predictions proving that the data used for the screening are in their normal state thus, there won't be need for data transformation or selection of a different one. Mathematically

Jarque-Bera test is computed as: $JB = \frac{n}{6} (s^2 + \frac{(K-3)^2}{4})$ Where: n stands for data population, S means skewness, K stands for kurtosis. The outcome expected from the computation is one that the JB test will have a distribution of two distinct degrees of freedom signifying a normal

distribution. While on the flip side, if the result of the computation is different from the distribution, the null hypothesis is rejected, signifying that the data collected for the purpose of the inquiry is not fit to be used for the projections. This is determined using a P-value, usually at a 5% level of significance.

3.7.2 Serial Correlation Test

This test examined relationship that existed between the variables current value and its past values. This test is commonly known as autocorrelation test, this test proves its usefulness in determining future value, it explored the assumption of independence by detecting any form of violation. This test also helps to detect problems associated with Ordinary Least Squares (OLS) estimation and hypothesis testing. After the autocorrelation testing, it is expected that the result would come out as a positive serial correlation which means that the values of the variable in question has the tendency to move in the same direction in the future as in the past or a negative serial correction, which implies that the values are likely going to move in opposite direction as they have had in the past. If the serial correlation is zero, it simply translates to no correlation with independent observations. There are several types of serial correlation tests and these includes: Durbin and Watson test, Wooldridge test, and Breusch and Godfrey test. This inquiry as against other type of the tests, applied the mechanism as crafted by Trevor Breusch and Leslie Godfrey (1978), because it has proven to be useful for technical analysis coupled with robustness as it gives room for higher orders of serial correlation beyond compatible lag periods. At the end of the assessment, the decision was made based on the threshold of 5% level of statistical significance to notify if there is a serial correlation. The formula for serial correlation based on Durbin-Watson's test as was used in this study is expressed as: $d =$

1 $\frac{\sum_{t=2}^T (\varepsilon_t - \varepsilon_{t-1})}{\sum_{t=1}^T \varepsilon_t^2}$ where: d stands for Durbin-Watson statistic, ε_t represents residual, and T stands
2 for number of observations.

3

4 **3.7.3 Heteroscedasticity Test**

5 This test centered on the residuals to determine whether there was heterogeneity of the variance,
6 based on the assumption of Ordinary Least Square (OLS) that residuals must have a constant
7 variance. The test of heteroscedasticity will show if the error terms are normally distributed.
8 Heteroscedasticity may occur in a situation where there is misspecification of the model, nature
9 of the variables, outliers, and incorrect functional form. This research utilized the crafted by
10 Breusch and Pagan (1979), the mechanism played a crucial role in the detection of
11 heteroscedasticity, because it is evidently sensitive to the assumption of normal distribution of
12 residuals. Its application is important for time-series data since error terms often exhibit
13 heteroscedasticity when regressors change significantly across the series. The Breusch-pagan
14 test for heteroscedasticity is expressed as:

15 $H_0 : Var(u|x_1x_2, \dots, x_j) = Var(u|x) = \sigma^2$ where: $u|x$ stands for residuals of the
16 parameters in the inquiry and σ^2 denotes residual's variance. At the decision point of this
17 mechanism, the inquiry checked to ensure that the residuals are normally distributed of which
18 if the case is reverse then, this indicates the presence of heteroscedasticity.

19

20 **3.8 Justification of the Methods Used**

21 The adoption of the Autoregressive Distributed Lag mechanism as the technique for examining
22 how the selected economic factors of this inquiry played a part in the demise of new born babies
23 in the Nigerian state from 1990–2024 is justified due to its unique ability to screen deep into

1 far-reaching periods to capture how these indicators connect. According to what was postulated
2 by Pesaran *et al.* (2001), the mechanism comes with many components for examples, the
3 inquiry was able to check for cointegration at levels and at first difference without any
4 significant complications, thereby leading to accurate and straightforward results. Additionally,
5 another the components of the technique was the error correcting component, which was
6 responsible for detecting the speed of calibration to normalcy. This is interpreted in the inquiry
7 as the speed with which infant deaths adjusts to its current rate before distortions caused by
8 economic elements or simply the speed of adjustment to equilibrium following macroeconomic
9 shocks. It further ensures long-run stability by measuring how quickly economic and health
10 interventions influence IMR, the value that represents this phenomenon is expected to have a
11 negative number while the probability should maintain a significance at 5% threshold and this
12 indicates a far-reaching connection among the variables.

CHAPTER FOUR

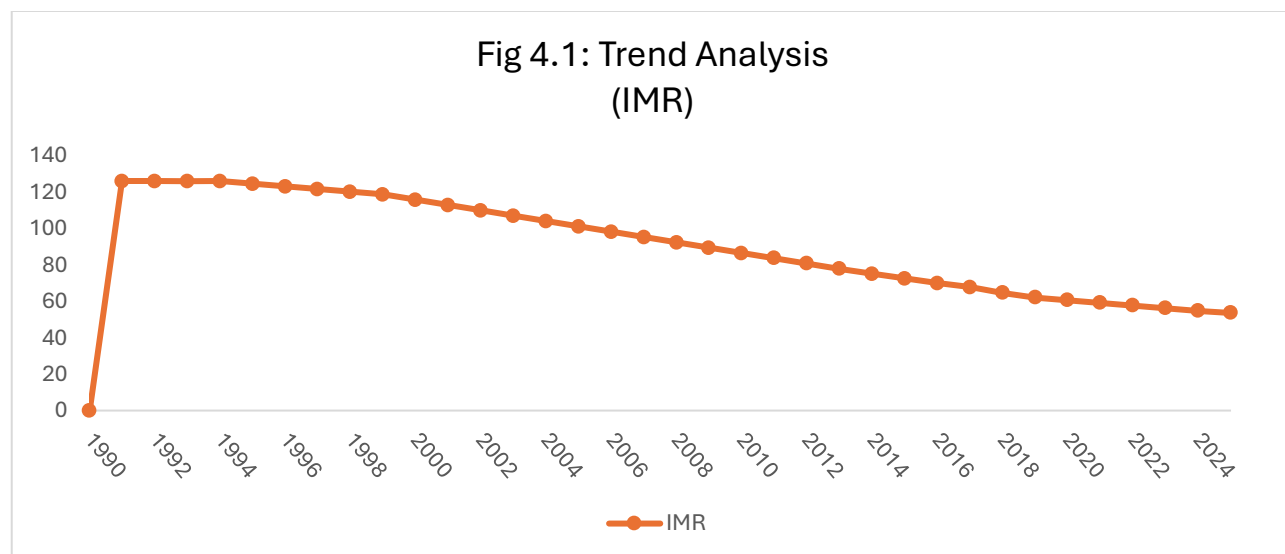
DATA PRESENTATION AND ANALYSIS OF RESULTS

To empirically investigate how these macroeconomic variables chosen in the inquiry to find out how they influence infant deaths in Nigeria. The model estimation was conducted using historical data that spanned from 1990 to 2024. The economic tools used to explain the persistent infant death are as follows: GDP Per Capita (GDPPC), Government Health Expenditure (GHE), Access to Electricity (ACE), Health-Specific Aid (AID), and Infant Mortality Rate (IMR). These variables were carefully selected to capture both economic performance and health-related macroeconomic factors that are theoretically connected to infant health outcomes. The data set provides a sturdy basis for the inquiry to examine the short-term influence and the far-stretched connection among the variables within the ARDL mechanism. The regression data used for the estimation is presented in Table 4.1 in the Appendices.

4.2 Data Analysis

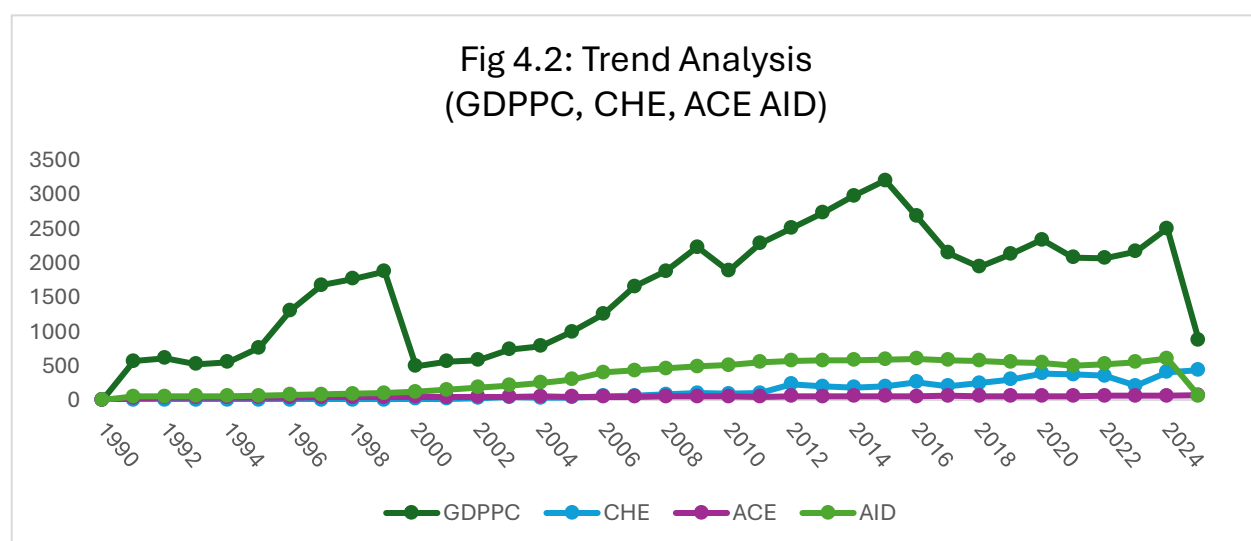
4.2.1 Trend Analysis

The agreement in the third phase of this inquiry is the execution of the analysis of the data which will begin with the analysis of the trends of the individual sets of data provided for the chosen indicators including rate of infant deaths. This is an important stage which will comprehensively display the behavior and the movement over time.



Source: *Author's Analysis with Excel (2025).*

As seen in the illustration Figure 4.1, infant mortality in Nigeria from 1990 to 2024 showed a steady and unwavering decline over the years. The highest level of infant deaths attained was from 1990-2000 whereas the lowest point started from 2020-2024 which is probably as the result of improvements in the healthcare access, maternal education, health AID, etc.



Source: *Author's Analysis with Excel (2025).*

1 Following the movements in the trend analysis as observed in fig. 4.2 captured trends for
2 national income per capita, capital health expenditure, Access to electricity, and Foreign Aid in
3 Nigeria. The GDPPC showed fluctuations but apparently the movement is upward in general
4 and likely affected by change in the price of crude oil. Upward periods are due to economic
5 instabilities. Also, for the trend of capital health expenditure by Nigeria's Government showed
6 consistency in growth as a result of governments commitment in investing in the health sector
7 although this spending is yet not sufficient to improve relevant infrastructure to world standard.
8 Meanwhile, the trend in fig. 4.2 for access to electricity (ACE) evidently shows the electricity
9 instability problem in Nigeria There was apparent upward trend in access to electricity in
10 Nigeria from 1990-2024. This improvement was due to inevitable urban expansion, energy
11 improvement programs, and investment in the power sector. Power outages and instabilities
12 still ravage the nation despite crucial intervention by the government. The trend on the foreign
13 aid (AID) translates to Nigeria's reliance on foreign support, due to the nation's need for growth
14 in the health sector and in the economy at large. This is clearly seen in that chart as the indicator
15 demonstrated mostly an upward movement.

4.2.2 Descriptive Analysis

Descriptive analysis was computed to give a meaningful summary to each indicator focused on in the inquiry. The basic diagnostics starts with the tests as follows: mean, median, maximum, minimum and standard deviation.

Descriptive Analysis and Summary Statistic of the Variables

Table 4.2: Descriptive Analysis Table

	IMR	GDPPC	CHE	ACE	AID
Mean	91.94497	1636.231	132.4343	48.70229	344.5429
Median	92.25500	1872.000	81.91000	49.97000	430.0000
Maximum	125.8740	3201.000	434.7900	70.00000	604.0000
Minimum	53.70000	494.0000	0.150000	27.30000	50.00000
Std. Dev.	25.40127	812.6397	139.0126	9.075104	221.1146
Skewness	-0.050406	-0.013917	0.782693	-0.095740	-0.213676
Kurtosis	1.546022	1.784116	2.292091	2.755247	1.280926
Jarque-Bera	3.097815	2.157093	4.304375	0.140830	4.576022
Probability	0.212480	0.340089	0.116230	0.932007	0.101468
Sum	3218.074	57268.07	4635.200	1704.580	12059.00
Sum Sq. Dev.	21937.63	22453030	657033.2	2800.156	1662317.
Observations	35	35	35	35	35

Source: *Author's Evaluation with EViews 12 (2025).*

In the inquiry, one of the analyses carried out was the description of the data for the respective indicator and was reflected by the standard deviation, mean, kurtosis, and skewness, as presented in Table 4.1. The average of Infant Mortality Rate is 93.06 and ranges from 54.74-125.87 while skewness and kurtosis -0.08 and 1.56 respectively. GDPPC recorded an average of 1658.6, which ranges between 494 and 3201 with skewness and kurtosis of -0.07 and 1.81. CHE showed an average of 132.4 with the ranges between 0.15 and 434.8, with skewness and kurtosis of 3.87 and 16.57. Also, Access to electricity recorded an average of 48 with skewness and kurtosis of -0.36 and 2.43 while AID recorded an average of 344.5, which ranges between 50 and 604 with skewness and kurtosis of -0.214 and 1.28.

4.2.3 Correlation Analysis

Table 4.3 Correlation Analysis

Corr. Probability	IMR	GDPPC	CHE	ACE	AID
IMR	1.000000	-0.699859	-0.677323	-0.934300	-0.828272
GDPPC	-0.699859	1.000000	0.2302660	0.597007	0.843729
CHE	-0.677323	0.2302660	1.000000	0.7054473	0.273272
ACE	-0.934300	0.597007	0.705447	1.000000	0.711442
AID	-0.828272	0.843729	0.273272	0.711442	1.000000

Source: Author's Evaluation with EViews 12 (2025).

As proposed in chapter 3, the correlation analysis was conducted with coefficients expected to fall between +1 and -1. A value of +1 indicates that the magnitude of the association among the variables is perfect positive relationship while a value lesser than zero means a negative correlation, and if the value is naught, it means there is no correlation. Outcome from the computation as seen in table 4.3 based on the values of the correlation of the selected economic indicators as against infant mortality showed a negative correlation, this outcome aligns with the a priori expectations, as higher socioeconomic development, improved government spendings, foreign aids are associated with reduction with infant deaths.

4.3 pre-estimation Test

4.3.1 Unit Root Test

Another important test necessary in this phase is the stationarity test and will be possible by using the Augmented Dickey-Fuller test to find out the status of the available data.

Table 4.4: Unit Root Test Result

variables	ADF Values	Critical Values	Order
IMR	-2.938997*	-2.636901	I (0)
GDPPC	-4.273570*	-2.636901	I (1)
CHE	-3.905995**	-3.587527	I (1)
ACE	-2.790636*	-2.636901	I (0)
AID	-2.226164**	-1.951332	I (1)

* is for 1% significant level, ** for 5%, and *** signifies 10%.

1 **Source:** *Author's Evaluation with EViews 12 (2025).*

2 The result of this computation at this point in the inquiry as shown in Table 4.4. indicated that
3 infant mortality rate was integrated at order zero showing no sign of unit root. While for the
4 chosen economic indicators, GDPPC, CHE and AID were found to be stationary at the first
5 difference but only ACE came up at levels. Following these outcomes and the respective order
6 of integration as exhibited by these indicators, the long run connection or expectation was
7 projected using the ARDL framework. The framework has proven to show buoyancy when
8 alternating between unit root at levels and first difference (Pesaran *et al.* 2001).

9

10 **4.3.2 Co-integration of ARDL-Bounds Test**

11 This section shows the ARDL co-integration bounds test result of the variables.

Table 4.5: ARDL-Bounds Testing

Null Hypothesis: No long-run relationship exists

Test Statistic	Value	Significance	I (0)	I (1)
F-statistic	162.7671	10%	2.2	3.09
k	4	5%	2.56	3.49
		2.5%	2.88	3.87
		1%	3.29	4.37

Source: *Author's Evaluation with EViews 12 (2025).*

12 This evaluation being one of the components of the ARDL framework in table 4.5 shows how
13 the selected economic indicators connect with infant deaths in the long run the ARDL bounds
14 test for co-integration was carried out for all models based on the research objectives. The F-
15 statistics value of 162.7671 appeared greater than critical values at all bounds. The conclusion
16 given by the inquiry per the result implies that the chosen macroeconomic indicators showed
17 connections to the deaths of new babies in long term.

18

19

4.4 Model Estimation Result

This part of the phase four of the inquiry centers on the regression of the data provided for selected macroeconomic proxies namely: GDP per capita (GDPPC), capital government health expenditure (CHE), access to electricity (ACE) and health-specific aid (AID), against the rate of infant deaths in Nigeria. it shows the short- and long-term expectations at the end.

Table 4.6.1: ARDL Regression Results		Dependent Variable: IMR		
Variable	Coefficient	Std. Error	t-Statistic	Prob.*
IMR (-1)	0.893954	0.044039	20.29891	0.0000
GDPPC	0.000592	0.000142	4.159451	0.0013
GDPPC (-1)	-5.93E-05	0.000170	-0.349357	0.7329
GDPPC (-2)	1.14E-05	0.000155	0.073032	0.9430
GDPPC (-3)	4.93E-05	0.000168	0.293812	0.7739
GDPPC (-4)	-0.000328	0.000133	-2.472568	0.0294
CHE	-0.001133	0.001284	-0.882609	0.3948
CHE (-1)	0.001448	0.001486	0.974765	0.3489
CHE (-2)	-0.001865	0.001669	-1.117510	0.2857
CHE (-3)	-0.001399	0.001690	-0.827523	0.4241
CHE (-4)	0.004789	0.001403	3.413318	0.0051
ACE	-0.040936	0.017438	-2.347477	0.0369
ACE (-1)	-0.077347	0.019766	-3.913185	0.0021
ACE (-2)	-0.046867	0.021770	-2.152840	0.0524
ACE (-3)	-0.014440	0.021490	-0.671929	0.5144
ACE (-4)	-0.037625	0.017690	-2.126892	0.0549
AID	-0.002832	0.000841	-3.365440	0.0056
AID (-1)	-0.003434	0.001667	-2.059901	0.0618
Coint Eq (-1) *	-0.106046	0.002851	-37.19570	0.0000
R-squared: 0.999975, Adj. R-squared: 0.999937, D&W stats: 2.949797, F-stat: 26327.80, and Prob: 0.000000.				

Source: Author's Evaluation with EViews 12 (2025).

From Table 4.6, the outcome revealed that F-statistic value showed 26327.80 while the probability value recorded 0.0000 indicating that the selected macroeconomic indicators is connected to the demise of new born babies in long term in Nigeria. The one-period ECT (-1) value of -0.106046 was found to be highly significant at 1%. This implies that there is a case of disequilibrium in the Infant Mortality Rate model, it would take a speed of about 10% to reach

adjustment to equilibrium from short-run to the long-run. Furthermore, the Adjusted R-squared of 0.999937 means that the selected macroeconomic variables accounted for about 99% change in the Infant Mortality Rate, while the remaining 0.000063% accounts for other factors affecting the dependent variable not captured. This result demonstrates a strong explanatory power and a good overall fit of the model.

Table 4.6.2: ARDL Extension, Long Run Results.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GDPPC	0.002508	0.001386	1.808857	0.0956
CHE	0.017357	0.045567	0.380902	0.7099
ACE	-2.048302	0.512111	-3.999721	0.0018
AID	-0.059087	0.013541	-4.363389	0.0009

Source: *Author's Evaluation with EViews 12 (2025).*

The results in Table 4.6.2 revealed that GDP per capita demonstrated a positive and statistically significant relationship with infant mortality in the long-run while capital health expenditure appeared positive but statistically insignificant in the long-run, indicating that increased capital allocation to health may not have translated effectively into measurable improvements in infant mortality. Access to electricity maintained a negative but statistically significant connection to the demise rate of infants, implying that improved access to electricity enhances healthcare delivery and household welfare, thereby reducing infant deaths. Also, the health-specific aid maintained a similar stance with the death rate of infants which is a negative but significant connection in a long term, highlighting that foreign aid targeted at the health sector was important in lowering infant mortality in Nigeria.

4.5 Post Estimation Test Results

Post-Diagnostic Checks

Table 4.7: Results of Post-Diagnostic Checks

Test		Outcomes	
		Coefficients	Probability
Serial Correlation Test	F-stats	1.527989	0.2822
Heteroskedasticity:	F-stats	0.894580	0.5964
Normality Test	Jarque-Bera	1.972475	0.372977

Source: *Author's Evaluation with EViews 12 (2025).*

The post diagnosis as applied in the inquiry translates to the following: the first result in the result table shows that the F-statistics came out as 1.53 and the probability made this result usable as it exceeded the 5% threshold. The inquiry would carry on since there was no serial correlation in this framework that seeks to discover how the chosen indicators influence death rate of infants in Nigeria. Secondly, the heteroskedasticity results show that variables are free from the problem of Heteroskedasticity. As per the result generated F-statistics value in computation came out as 0.89 with a probability value of 0.596 which exceeded the 5% threshold. This implies the absence of heteroskedasticity in the model of how macroeconomic variables connect to the rate of demise of infants in Nigeria. And finally, the Jarque-Bera test of normality shows that the error term in our specified equation is normally distributed. Finally, this is seen as the respective insignificant Jarque-Bera statistics of 1.97 and the probability value of 0.37.

4.6 Test of Research Hypotheses

4.6.1 Hypothesis One Result

Table 4.8: Wald Test for Hypothesis 1- GDP Per Capita and Infant Mortality

Test Statistic	Value	Diff.	Probability
F-statistic	184.4389	(4, 12)	0.0000
Chi-square	737.7556	4	0.0000

Source: *Author's Computation Using EViews 12 (2025).*

1 **H₀**: There is no significant impact of GDP per capita on infant mortality in Nigeria.

2 **H₀₁**: There exists a significant impact of GDP per capita on infant mortality in Nigeria.

3 To address the first research question, “To what degree would new born death rates reach given

4 the influence by income per individual in Nigeria?” and simultaneously fulfilling the

5 corresponding objective of investigating how income per individual impact newborn death rates

6 in Nigeria, the ARDL framework was employed. A Wald test was applied, and the outcome put

7 out a p-value of 0.0000, which is lesser than the benchmark 5% level of significance.

8 Consequently, the decision of this inquiry was the rejection of null hypothesis and accept the

9 alternative hypothesis. This result implies that there exists a significant connection among

10 income per head and the deaths of infants in Nigeria.

4.6.2 Hypothesis Two Result

Table 4.9: Wald Test for Hypothesis 2- Government health exp. and Infant Mortality

Test Statistic	Value	Diff.	Probability
F-statistic	4.270960	(4, 12)	0.0223
Chi-square	17.08384	4	0.0019

11 **Source:** *Author’s Evaluation with EViews 12 (2025).*

12 **H₀**: Government health spending does not significantly influence infant mortality in Nigeria.

13 **H₀₂**: Government health spending significantly influences infant deaths in Nigeria.

14 The solution to second research question “How does spending on health by relevant authority

15 impact new born deaths in Nigeria?” and corresponding objective, “To Examine the impact of

16 health spending by the relevant authority on infant mortality in the country” the ARDL

17 framework was employed. The Wald test result, as displayed in the platform disclosed that funds

18 allocated by government for health influences infant mortality significantly. This is because the

19 p-value recorded 0.0223 and is lesser than 0.05 at 5% level of significance. This leads to the

1 rejection of the null hypothesis and accept the alternative which implies that there exists a
 2 significant impact of government health expenditure on infant mortality in Nigeria.

4.6.3 Hypothesis Three Result

Table 4.10: Wald Test for Hypothesis 3- Access to Electricity and Infant Mortality

Test Statistic	Value	Diff.	Probability
F-statistic	4.913896	(4, 12)	0.0140
Chi-square	19.65559	4	0.0006

3 **Source:** *Author's Evaluation with EViews 12 (2025).*

4 **H₀:** Access to Electricity has no significant impact on infant mortality in Nigeria.

5 **H₀₃:** Access to Electricity has a significant impact on infant mortality in Nigeria.

6 To answer the third research question, “In what ways does access to electricity impact infant
 7 mortality in Nigeria?” and to fulfill the objective, “To analyze the contribution of access to
 8 electricity in mitigating infant mortality in Nigeria,” the Wald test within the ARDL framework
 9 was applied. Also, for the third, the computation showed a probability of 0.0140 and this implies
 10 significance within the benchmark. Thus, the decision by the inquiry was accept alternative
 11 hypothesis. This outcome confirms that access to electricity exerts a considerable amount of
 12 influence on infant death rate in Nigeria.

4.6.4 Hypothesis Four Result

Table 4.10: Wald Test for Hypothesis 3- Access to Electricity and Infant Mortality

Test Statistic	Value	Diff.	Probability
F-statistic	15.31302	(1, 12)	0.0021
Chi-square	15.31302	1	0.0001

13 **Source:** *Author's Evaluation with EViews 12 (2025).*

14 **H₀:** Health-Specific Aid has no significant influence on death rates of infants in Nigeria.

15 **H₀₄:** Health-Specific Aid has a significant impact on infant mortality in Nigeria.

16 To address the fourth research question, “What impact does health-specific aid have on reducing
 17 infant mortality in Nigeria?” and to achieve the objective, “To Evaluate the impact of health-

specific aid on improving infant mortality outcomes in Nigeria,” the Wald test within the ARDL framework was conducted. The test of this hypothesis showed a probability of 0.0021, which is well below the bench mark level of 0.05. The value gave this inquiry a reasonable ground in the decision to accept alternative hypothesis. On this note, the inquiry established that health specific aid exerts a strong connection to the demise of infants in Nigeria.

4.7 Discussion of Findings

This inquiry relied on the discovery, post application of the ARDL technique to establish the expectations among the respective macroeconomic tools and infant death rate in Nigeria. Based on the discovery, in the short-term case, the tool exerted a strong influence for better outcomes in new babies' health (see Table 4.6.1). This short-run outcome is consistent with Solow growth model, which noted that brief distortions in policies often produce quick but limited effects on social indicators. The extended term results showed mixed attributes, and the interpretations are presented according to the study objectives. For the first specific target which was to ascertain the impact of GDP per Capita on Infant Mortality. The outcome turned out as GDP per capita having a statistically significant and positive relationship with infant mortality (coef. = 0.002508; $p = 0.0956$; see table 4.6.2). What this result implies is that in the long run, increases in GDP per capita may not necessarily translate into reductions in infant mortality, highlighting the possibility that economic growth alone may not be crucial in driving improvements in child health outcomes. This supports the argument in Solow-swan that income gains without equal spread among citizens do not automatically improve child survival. Between Government Health Expenditure and Infant Mortality, the outcome came in as capital government health expenditure recording a positive coefficient and statistically insignificant (coef. = 0.018636; p

1 = 0.8964; Table 4.6.2). Interestingly, this contradicts the a priori expectation of a negative
2 relationship. This unexpected result suspects the usual problem of inefficiencies in the health
3 fund management of the health sector resulting to lack of improvement of infant health
4 outcomes. This finding aligns with Akinlo and Sulola (2019), where the findings showed that
5 capital expenditures in health sectors of developing nations such as Nigeria face implementation
6 roadblocks, e.g., poor implementation and corruption. This contradiction corresponds with
7 framework of Solow growth theory, which warns that weak institutions can nullify the expected
8 impact of government spending. For the evaluation among Access to Electricity and Infant
9 Mortality, the output showed that Access to electricity connected negatively and substantially
10 with infant death rate (coef. = -2.048302 ; $p = 0.0018$). what this result suggests is that improved
11 access to stable electricity supply can contribute to reducing infant deaths, especially when
12 coupled with appropriate policy measures that strengthen healthcare delivery systems in
13 Nigeria. This outcome agrees with the finding of Mohammed and Akuoko (2022), who in their
14 study argued that access to reliable electricity coupled with health infrastructure development
15 policies are crucial in the aim to address infant mortality issue. This also aligns with the Solow–
16 Swan model, where technological advancement including electricity enhances productivity and
17 improves well-being in the area of health. lastly, in the assessment of Health-Specific Aid and
18 Infant Mortality linkages, the result sprouted Health-specific aid exhibiting a negative and
19 statistically significant relationship with infant mortality (coef. = -0.188402 ; $p = 0.0001$; Table
20 4.6.2). This negative relationship with infant mortality corresponded with the a priori
21 expectation and as well aligns with Doucouliagos *et al.* (2021). The decision of the inquiry with
22 regards to health aid is that the indicator particularly reduces the demise rate of infants.
23 However, it is dependent on the quality of governance. This supports Human capital theory,

1 which emphasizes that external health funding yields stronger outcomes when institutional
2 quality is high. The error correcting term, which is the calibration mechanism, came out as -
3 0.106046 with a probability of 0.0000, which further means that these indicators connect in
4 longer periods. The term is approximately -0.11, and this means that 11% of any economic
5 distortion or shock, the calibration to equilibrium occurs each year. With this evidence, the
6 federal government of Nigeria is tasked with deploying consistent and sustainable policies over
7 time because an insignificant portion of imbalances corrects quickly. Therefore, long-term
8 planning and strong policy commitment are critical to maintain progress in reducing infant
9 mortality.

11 **4.8 Policy Implications of Findings**

12 In the short-term case of this inquiry, the discovery is that the respective indicators examined
13 exerted a positive but significant influence on death rate of infants in Nigeria. The error
14 correcting term which is the calibration mechanism came out as -0.106046 with a probability
15 of 0.0000 further means that these indicators connect in longer periods. The term is
16 approximately -0.11, and this means that 11% of any economic distortion or shock, the
17 calibration to equilibrium occurs each year. With this evidence, federal government of Nigeria
18 is tasked with deploying consistent and sustainable policies over time which in recent times
19 have been initiated in form of Maternal and Newborn Mortality Reduction Initiative (MAMII)
20 and the Nigeria Child Survival Action Plan (NCSAP), because insignificant portion of
21 imbalances correct quickly. Therefore, long-term planning and strong policy commitment are
22 critical to maintain progress in reducing infant mortality. For the policy makers, there is need
23 for implementation of fiscal and monetary policy, which can be managed to stimulate stable

1 macroeconomic state through careful government spending, reduced unplanned policy change
2 and inconsistencies, and inflation control monetary stance that prevent economic distortions
3 which may worsen infant health expectations.

4 The far-stretched period results translate that for income per head (GDPPC) the value popped
5 out 0.002508 with probability of 0.0956. What the inquiry explained here is that when economic
6 grows the income as spread among citizens in the nation which improves spending powers tend
7 to slightly reduce the deaths of infants in extended periods. The less substantial nature of this
8 connection points to the importance of inclusive growth policies. The policy implication is that
9 the Nigerian government should prioritize job creation, poverty alleviation, and equitable
10 income distribution, as these measures can directly improve household welfare, improve access
11 to healthcare, and ultimately reduce infant deaths. With careful implementation of fiscal policy
12 such as the National Employment Policy (NEP) 2025 and the Progressive Tax Reforms
13 Initiative which would bring about increased spending, social protection programs, and funding
14 health services in underserved areas, while monetary policy can stabilize purchasing power by
15 curbing inflation to increase purchasing power of households to enable them afford healthcare
16 and nutrition needed for infant survival.

17 Contrary to this preceding outcome, capital government health expenditure (CHE) with the
18 values: coefficient of 0.017357 and probability of 0.7099 translate as positive but statistically
19 insignificant outcome which implies that an increase capital health spending alone may not
20 guarantee reduced infant mortality. thus, the federal government is tasked to invest more
21 concerns to providing efficiency; supervision committee for accountability and to curb
22 corruption; quality of health infrastructure investments; ensuring funds translate to accessible
23 and effective health services. Fiscal policy can directly strengthen this by improving public

1 financial management systems, ensuring transparent budgeting, and ring-fencing health
2 allocations. Monetary authorities, by stabilizing macroeconomic conditions, can prevent budget
3 erosion caused by inflation or exchange-rate shocks that undermine health-sector investment
4 efficiency. Furthermore, access to electricity recorded a Coefficient of -2.048302 and a p-value
5 of 0.0018, proving Significantly negative. This agrees with the craft of the Solow and Swan
6 (1956), where technological advancement contributes to economic output growth. Hence, the
7 federal government should focus on the improvement in electricity access while expanding
8 reliable electricity access should be a health policy priority. It is also essential for the federal
9 government to investment in rural electrification, power grid reliability, and renewable energy
10 which is crucial and could directly improve health facility operations and living conditions
11 while reducing child deaths. with the enhancement of fiscal policy, these measures can be
12 supplemented by capital investment, tax incentives for renewable energy, and expansion of
13 public and private partnerships, while monetary policy can reduce borrowing costs for power
14 sector investments by maintaining encouraging interest rates. Lastly, the health-specific aid
15 values such as: coefficient of -0.059087 and probability of 0.0009 which implied a significant
16 negative, meaning that more health aid reduces infant mortality. Therefore, the study tasked
17 Federal government and foreign donors with maintaining and expanding health aid programs
18 while focusing policies on effective coordination, transparency, and maximized spending to
19 ensure reduction of infant deaths. Fiscal policy can enhance this by ensuring foreign funding
20 and efficient allocation of AID projects, while monetary policy ensures friendly exchange-rate
21 so that the real value of foreign health aid is not diminished allowing such interventions to
22 remain useful.

23

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Major Findings

The idea behind this inquiry was to discover how the macroeconomics selected connects to the changes in the rate at which infants die in Nigeria utilizing a time series from 1990–2024. Major findings are summarized hereunder:

The first objective was to examine how income per head (GDPPC) influences infant mortality in Nigeria. In the short-run, the result mainly showed income per head to have significant and beneficial short-run effects. Furthermore, the results in longer period model showed the income per head exhibited a positive and marginally significant effect, suggesting that as economic activities expand it leads to a fall in the death rate of infants, its impact remains weak and insufficient in isolation. This finding points to the need for policies that transform economic gains into health improvements through inclusive growth, poverty alleviation, and partial access to health care.

The objective number two was to determine how capital government health expenditure affects infant mortality. Based on the analysis, some lags of capital health expenditure also showed significant effects in the short run. Also, the capital health expenditure was positive and insignificant and suggests poor spending efficiency. Generally, the capital health spending needs better management to be effective.

For the third objective, it aimed to assess the impact of access to electricity on infant mortality. Access to electricity also appeared negative and highly significant, which is indicative to strong reduction in infant mortality with better access. Improved electricity access is therefore considered effective in reducing infant mortality in both short and longer periods.

1 Lastly, the objective number four targeted to evaluate the role of health-specific aid in reducing
2 infant mortality. In the analysis, Health-specific aid also appeared negative and highly
3 significant and evidently proves that aid is crucial and effective for infant deaths eradication.
4 Health-specific aid is therefore one of the most effective indicators in the model.
5 For the Error-Correction Adjustment (ECM) and Overall Interpretation, the error-correction
6 mechanism recorded negative and statistically significant. Thereby, confirming that in longer
7 period the core variables as utilized showed equilibrating connection. However, the relatively
8 slow adjustment rate of approximately 11% correction annually indicates that deviations from
9 equilibrium take considerable time to revert to normal. Hence, the federal government is tasked
10 with the need for consistency and sustained health policy efforts.

12 **5.2 Conclusion**

13 The idea behind this inquiry was to discover how the selected economic tools namely - GDP
14 per capita, capital government health expenditure, electricity access, and health-specific aid on
15 infant mortality in Nigeria from 1990 to 2024. ARDL results per longer periods revealed that
16 income per head exerts a positive influence in reducing death of infants. This influence is only
17 marginally significant, suggesting that growth alone may not be enough to deliver sustained
18 improvements on infant health. The Capital government health expenditure displayed a strong
19 connection in longer period, indicating weak supervision, inefficiencies, and poor utilization of
20 health infrastructure investments. On the contrary, access to electricity and health-specific aid
21 were seen to be statistically significant with a negative strong relationship with infant mortality
22 given their level of significance at under 1%, which proves their indispensability in the
23 reduction of infant deaths to barest minimum.

1 The short-run ECM results further supported these findings, as all selected macroeconomic
2 variables showed significant and immediate impacts in reducing infant mortality. Some lags of
3 capital health expenditure also appeared significant, highlighting the importance of timely and
4 efficient spending. The value representing calibrating mechanism was negative and reliable in
5 statistic terms, confirming stability in the longer period connection. The speed of calibration to
6 normalcy points to gradual convergence toward equilibrium. With what has been deduced from
7 these findings, it is noteworthy that reducing infant mortality in Nigeria is systemic and requires
8 more than macroeconomic growth alone. There is a clear need for policies that ensure effective,
9 efficient, and well-planned health spending, reliable access to electricity targeting vulnerable
10 demography which are mainly rural and underserved areas, and enhance the management and
11 impact of health-specific aid. A well-planned health sector development programs and
12 sustainable policy approach will be essential to deliver lasting solution to premature infant
13 deaths which when dealt with would reinforce Nigeria's broader development goals.

15 **5.3 Recommendations**

16 This study highlights the critical role of energy access as a fundamental driver of health and
17 development, suggesting that targeted investments in this area can yield substantial long-term
18 economic and social benefits. According to the respective objectives, the recommendations are
19 outlined:

- 20 i. Stimulate Economic Growth in every region of the nation, based on the proof that
21 increased income per capita can significantly lower infant mortality rates, by strictly
22 and effectively aligning to the Renewed Hope Agenda for Fiscal and Monetary Stability,
23 the National Planning Commission and the Central Bank of Nigeria (CBN) should

1 ensure job creation and poverty alleviation for seamless access by household to
2 healthcare and basic needs, consistent with principles outlined in the Solow-Swan
3 Growth Theory.

4 ii. Promote effectiveness and efficiency in Health Spending. Due to the negligible impact
5 of current capital health expenditure, the Ministry of Health and the Budget Office must
6 implement stricter management to be certain that funds are channeled towards maternal
7 and child health services, with the aim of achieving measurable expectations.

8 iii. Increase reliable electricity supply for the optimal functioning of health facilities. Given
9 the significant effect of electrification on reducing infant mortality, the Ministry of
10 Power and the Rural Electrification Agency should expand the use of renewable energy,
11 particularly solar power, at primary health centers in underserved areas.

12 iv. Effective use and management of Health Aid. Considering that aid plays an important
13 role in lowering infant mortality, the Ministry of Foreign Affairs and the National
14 Planning Commission should ensure the transparent and organized application of
15 foreign support mainly for infants and maternal well-being in alignment with Nigeria's
16 health policy and the Sustainable Development Goal 3.

17 v. Ensure consistency and stability in the Policy for the Long Term. The gradual 11%
18 adjustment rate point to the need for consistent policy stance. The presidency and the
19 National Economic Council (NEC) should institutionalize a long-term strategy aimed
20 at reducing infant mortality, backed by sustained funding and continuous monitoring.

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5.4 Limitations of the Study

The major constraint encountered in this study was limited access to reliable and complete data for some of the selected economic tools. Also, the economic crunch was a thorn in the flesh in the financing, due to the exorbitant cost of materials needed to facilitate the study completion. The inquiry applied secondary data, comprising 35 adjusted annual observations. The reliability and completeness of the data, particularly for earlier years, may have affected the precision of the estimations while some relevant indicators were omitted as a result of inconsistencies or missing historical data. However, these limitations do not diminish the validity or policy relevance of the findings. The analysis utilized aggregate macroeconomic variables such as GDP per capita, inflation, and government health expenditure. While this provides a broad national perspective, it may mask heterogeneous effects across different population groups, regions, or sectors. For example, the impact of inflation or poverty on health outcomes could differ among income groups or states. The econometric techniques employed, including descriptive statistics, trend analysis, stationarity tests, and the ARDL model, are appropriate for analyzing linear relationships. However, these methods primarily capture linear effects and may not fully account for potential non-linearities, threshold effects, or reverse causality among the variables considered. Exploring these complex dynamics may require alternative methodologies that explicitly model such relationships. The focus of the inquiry was on the Nigerian economy. While this allows for an in-depth understanding of domestic macroeconomic and health policy linkages, it limits the comparability of the results to other countries with different economic structures, institutional settings, or health systems. Comparative studies involving similar emerging economies could enhance the external validity and applicability of the findings.

5.5 Contributions to Knowledge

This study aims to be of benefit to the federal ministry of health in formulating effective health policies linked to sustainable economic growth. It will also be beneficial to researchers, scholars, and students, particularly those of the economics department of Bingham University, in reaching the end of the tunnel of the gap in the Literature by explaining how government policies and programs can improve electricity access and how they subsequently influence health outcomes. In addition, the potential role of renewable energy solutions in reducing infant mortality in underserved or off-grid areas remains ignored. This research also incorporated four macroeconomic variables like: Gross Domestic Per Capita (GDPPC), Government Health Spending (GHE), Electricity Access (ACE), and Health-Specific Aid (AID) which was rarely examined together earlier studies typically considered only one or two, often excluding technological factors such as electricity access. Also, this inquiry will provide the latest analysis, based on the period from 1990 to 2024 and potential future research that seeks to address issues in health issues such as infant mortality and related areas.

5.6 Suggestions for Further Studies

Given the empirical investigation which analyzed the impact of selected macroeconomic variables (GDP per capita, capital government health expenditure, electricity access, and health-specific aid) on new born death rate in Nigeria, it is necessary to highlight certain gaps that remains unresolved. These gaps are critical in guiding future studies and addressing the persistent shortcomings in the health sector of Nigeria.

- 1 1. Further studies need to be conducted using more non-macroeconomic variables in areas
2 of social and demography such as - maternal education, access to clean water,
3 immunization, and cultural practices that predate infant mortality.
- 4 2. From what was gather by the observation on previous studies encountered by this paper,
5 future studies are implored to adopt alternative econometric methods or approach such
6 as panel data or spatial econometrics. They are also advised to try out other techniques
7 analysis such as vector error correction model and nonlinear autoregressive distributed
8 lag, to further expand the relationships shared by the variables of this study.

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APPENDIXES

2

Appendix A: Empirical summary

3

Table 2.1: Empirical Literature Summary Table

Author(s) & Year	Topic	Methodology	Findings	Gaps
Hashiani <i>et al.</i> (2015)	The effect of economic inequality on infant mortality.	Correlation Analysis Using STATA 12.	It was discovered that the rates of infant, under-5-year-old, and maternal mortality were high in undeveloped countries. given the economic indicators, per capita health expenditure, income per capita, and GNI were essential in creating disparities, whereas the out-of-pocket index had no impact.	Data were collected in the level of countries; consequently, the extension of its findings to individual level would lead to ecological fallacy. it was discovered also that covariates in creating inequality were not included in this study, rendering the decomposition of the inequality index and controlling for confounding and interaction effects impossible.
Heshmati (2018)	Causality between Gross Domestic Product and Health Care Expenditure in the Augmented Solow's Growth Model.	Regression Analysis and Causality Test.	The result showed a positive effect of healthcare expenditure on economic growth. Whereas the health expenditure model showed a close link between the variables characterizing the healthcare system of sample countries.	The use of causality in this paper may not capture factors like technology advancements, policy transition, and demographic shift due to complexities in the relationship between GDP and healthcare expenditure.
Attamah (2018)	Impact of government health	Vector error correction model (VECM).	This result showed a positive and significant impact of	Infant mortality rate was measured by only

Author(s) & Year	Topic	Methodology	Findings	Gaps
	expenditure on infant mortality rate.		the government expenditure on health on infant mortality.	government health expenditure and per capita income; this may not fully account for other important variables.
Ubi and Ndem (2019)	Poverty and health outcomes (life expectancy and infant mortality rate) in Nigeria.	Vector error correction model (VECM).	The result of this study indicates that poverty shock has no direct effect on health outcomes variable. Rather the study showed that variations in health outcomes are not actually due to poverty shocks but due to the health outcome shocks.	Given the time frame used in this study, it provides an opportunity for expansion to the timeseries and aids in the analysis of current trend to observe recent changes in the relationship between poverty and health outcomes such as infant mortality rate.
Akinlo and Sulola (2019)	The Effect of Energy use on Infant Mortality rates in Africa.	Panel Data Analysis.	It was found that other health determinants such as GDP per capita, health aid, HIV prevalence and immunization showed a significant negative effect on under-five and infant mortality.	This study focused on multiple countries and did not provide a detailed analysis on a single country or capture peculiar socio-economic and political determinants of public health expenditure and economic growth.
Shobande (2019)	The Effects of Energy use on Infant Mortality rates in Africa.	Gary Becker hypothesis and the Grossman model.	The study found that energy predictors showed a negative and significant impact on infant mortality rates in the African countries observed. however, results show	The study focus on energy use did not delve into the specifics of renewable and non-renewable energy sources and their

Author(s) & Year	Topic	Methodology	Findings	Gaps
			that a disturbing level of pollution causes an increase in mortality rates, whereas revenue gained from natural resource rents reduce the level of infant mortality.	respective impact on Infant Mortality Rate.
Ebeh <i>et al.</i> (2019)	Impact of infant mortality on economic growth in Nigeria.	Auto regressive Distributed Lag Model (ADRL) Technique.	The study discovered that infant mortality has significantly negative short run and long run impact on economic growth in Nigeria.	This study failed to account for grassroot differences. This because of regional inequality of health outcome and economic realities that is existential in Nigeria.
Raji (2020)	The role of macroeconomic factors on infant and under-five mortality rates in Nigeria.	Fully Modified OLS Technique.	The study revealed that the unfriendly macroeconomic policy focused on indicators like the instability of inflation rate, unemployment and exchange rate contributed to the worsened levels of infant and under-five mortality rate in Nigeria.	This study's recommendation lacks a detailed and sufficient strategies for policy makers in curtailing macroeconomic shocks on infant mortality rates in Nigeria.
Boachie <i>et al.</i> (2020)	The impact of health expenditure by the government in low-income and middle-income countries.	Fixed effects estimation technique and three-stage least squares.	The result highlighted the importance of public health expenditure as it is pivotal in the reduction of infant mortality. The result further revealed that as socioeconomic conditions such as access to clean water, income per capita, etc., health outcomes	The study lacked sufficient information on the difference in the government health spending and their various impact in the countries selected, regions and the grass root.

Author(s) & Year	Topic	Methodology	Findings	Gaps
			such as infant mortality improves.	
Karaman <i>et al.</i> (2020)	The impact of healthcare spending on health outcomes in OECD countries.	Stepwise Multiple Regression Analysis.	The result indicated that public healthcare spending per capita significantly and positively impacts maternal and infant mortality, male and female life expectancy at birth and at 80 years. it further revealed that private healthcare spending per capita was an important indicator of self-reported health.	This study did not account the effectiveness of health spending and the efficiency in the outcome of potential improved health care.
Kara and Ersin (2020)	The effect of health expenditure to decrease infant mortality rate in OECD countries.	The Kao panel cointegration and Dumitrescu-Hurlin causality analysis.	It was found that there is a positive long-term relationship between health expenditure and infant mortality rates in the selected countries.	This study focused only health spending ignoring other health macroeconomic variables that affect infant mortality rates in these countries.
Adeosun and Faboya (2020)	The effects of expenditure in healthcare on child mortality in Nigeria.	Vector Auto Regression Model Technique.	The proved that there exists a negative correlation between the healthcare expenditure by government and infant mortality which simply translates to spending on healthcare has no significant impact on infant mortality rate in Nigeria.	Looking at the technique employed in the analysis. The results found proves that the relationship among the chosen variables may not always be the same and this creates an avenue for further investigation using a different technique.
Adeleke <i>et al.</i> , (2022)	Spatial analysis of economic predictors of under-5	Spatial Statistics	The study proved that Nigeria has a high under-5 mortality rate	The study successfully identifies macroeconomic

Author(s) & Year	Topic	Methodology	Findings	Gaps
	mortality in Nigeria.		of 117 deaths per 1,000 live births.	predictors of under-five mortality, it may overlook other critical factors such as cultural, social, and healthcare system influences. these components have their respective influences on under-5 health but was not fully accounted for in his research
Rahman and Alam (2021)	The role of access to electricity, female education, and public health expenditure ⁴ with female health outcomes in the SAARC-ASEAN countries.	the corrected standard error (PCSE) model, and the Feasible generalized least square (FGLS) model	Panel error model The result showed that Access to electricity, female education rate, public health expenditure, economic growth, and immunization rate, all have a positive effect on female life expectancy at birth, and a negative effect on the female adult mortality rate, and also revealed a one-way causal relationship between variables.	There were no clear causal relationships due to the observational nature of the data collated.
Bugelli <i>et al.</i> (2021)	Health capabilities and the determinants of infant mortality in Brazil, 2004-2015: an innovative methodological framework.	Multilevel panel data with fixed effect nested within- clusters.	Following the implementation of these policies, the results also pointed out a relationship between employment rate and different infant mortality rates, noting that employment impacting neonatal mortality up to 3 years and that of a family income below 2	The focus on Brazil as a country has created an avenue for investigating other countries with similar trend to compare outcomes.

Author(s) & Year	Topic	Methodology	Findings	Gaps
			minimum wages means a high chance of infant deaths.	
Doucoulia <i>et al.</i> (2021)	The impact of health aid and governance on infant mortality.	Long Difference Estimator and Instrumental Variable Estimation.	The find of this study revealed that health Aid has high tendency to improve infant mortality however, it is dependent the quality of governance and the corruption eradication.	There is no clear distinction on the mechanism of governance in different system of governance and the efficiency of aid. This has created gap for future research.
Mohammed and Akuoko (2022)	Subnational variations in electricity access and infant mortality: evidence from Ghana.	Pooled Cross-Section Regression Approach.	The result of this examination established that fewer incident of infant mortality in regions with an improved access to electricity while on the contrary, the opposite is the case in the regions with epileptic power supply.	Unclear investigation into the disparity of access to electricity and the impact it has on infant mortality in urban areas as against rural areas.
Akpesiri and Eburajolo (2022)	An investigation of the monetary policy and the health sector with focal point on infant mortality outcome in Nigeria.	Vector Error Correction Model.	The result showed that literacy rate and Per Capita income exerted a negative effect on the infant mortality rate although they were statistically not significant.	The gap can be seen in the selected period of study. An adjustment onto the current years' observation can give a clear understanding of the chosen variables on infant mortality in Nigeria.
Logarajan <i>et al.</i> (2022)	The impact of public, private, and out-of-pocket health expenditures on under-5 mortality in Malaysia.	Autoregressive Distributed Lag (ARDL) Technique.	The result showed that out-of-pocket health expenditure affects the rate of under-five mortality whereas the public and private health expenditures are	Inconsistency of the published data on health expenditure. This calls for the provision of data on public and out of pocket health

Author(s) & Year	Topic	Methodology	Findings	Gaps
Aniefor (2022)	The effect of company income tax on infant mortality in Nigeria.	Regression Analysis.	statistically insignificant.in Malaysia. The result showed a negative and an insignificant effect of company income tax on infant mortality.	expenditure for future research. Company income tax as an independent variable may not fully account for variation in infant mortality without consideration of unemployment rate.
Adeosun et al. (2022)	Macroeconomic variables and mortality rate nexus: focus on Nigeria.	ARDL Technique	The result showed a positive relationship between GDP per capita and mortality rate on the short-run which means increase in GDP per capita does not reduce the mortality rate in the country, there is also a positive short-run relationship between inflation and mortality rate but there are no long-run relationships between the variables.	The period covered in this research may not be enough to cover the long-term effect of the selected variables and has created an avenue to future studies to uncover these long-term effects.
Bao <i>et al.</i> (2022)	The impact of real estate prices, inflation, and health outcomes: evidence from advanced economies.	2 stage least square (2SLS) and generalized method of moments (GMM)	The result showed a positive and significant effect of housing rent on infant mortality rate. It was also found that an increased inflation positively affects the infant mortality rate. While GDP and health expenditure has the tendency to improve health by reducing the infant mortality rate.	The used of the technique 2 stage least square (2SLS) and generalized method of moments (GMM) may not be accurate in the study. Thus, gives room for other techniques for the measure of human health.

Author(s) & Year	Topic	Methodology	Findings	Gaps
Jiang and Liu (2022)	This study examined the causal relationship between inflation and infant mortality in China.	Mixed frequency vector autoregressive model (MF-VAR)	There is a causal relationship exists between inflation and infant mortality. Notably, the Consumer price index has stronger explanatory power for infant mortality rate in mixed-frequency data which in national realities explains why it is important to maintain price stability.	This study needs broader analysis using more variables to fully understand the variations in infant mortality. This implies that the independent variable cannot fully explain these variations.
Asghar <i>et al.</i> (2023)	The impact of rural and urban electrification and biomass energy consumption on infant mortality rate.	Panel Quantile Regression (PQR) approach.	It revealed that in developed and developing countries, biomass energy consumption has a positive impact on infant mortality rate, whereas rural and urban electrification has shown an inverted U-shaped relationship with infant mortality in different quantile groups.	The Panel Quantile Regression (PQR) approach does not establish the long-run impacts and it's an opportunity to explore long-run impacts with relevant techniques.
Hoque and Rana (2023)	Examination of the impact of Social and Economic Indicators on Infant Mortality Rate in Bangladesh, India.	Vector Error Correction Model Technique.	This study showed that GDP, expenditure on education, and female labor force participation are observed to impact infant mortality rates negatively. Whereas the unemployment rate and female school enrollment were revealed to improve infant mortality rates.	Lack of comparison with similar countries to show how it is applicable to global standards.
Miladinov (2023),	Measuring of the socio-economic causes of infant mortality in Macedonia, Turkey and Albania.	Seemingly Unrelated Regression (SUR) method.	The result disclosed that the infant mortality rates drastically dwindled in all three countries throughout the observed period from	The research lacked the use of broader variables such as inflation, health expenditure, and access to

Author(s) & Year	Topic	Methodology	Findings	Gaps
			1991–2017, and a significant negative effect of these socioeconomic causes on infant mortality rate was also discovered.	electricity which may give a significant contribution to the changes in the dependent variable.
Koginam and Anthony (2023)	Economic growth and dimensions of mortality in Nigeria.	Autoregressive Distributed Lag (ARDL) approach.	The findings disclosed that infant mortality exhibits a negative but statistically insignificant on RGDP in the long run. However, infant mortality negatively and significantly influences RGDP in the short run.	There is an opportunity for future research to focus on the contrast relationship between infant mortality and economic growth to understand the dynamics of the variables.
Awoyemi <i>et al.</i> (2023)	A time series analysis of government expenditure and health outcomes in Nigeria.	Autoregressive Distribution Lag Technique.	This study revealed a negative relationship between health expenditure and mortality rate.	The theoretical framework does not account for technological advancement. It creates room for future examination with relevant theories.
Udemba <i>et al.</i> (2023)	Environmental implication of energy policies and private and public subsidies on infant mortality rate: a sustainable development study of India.	Relevant scientific methods residual Augmented Least Squares (RALS).	The study reveals that energy policies and subsidies targeting public and private sectors significantly impact infant mortality in India.	There is a gap regarding the geographical difference, creating further research opportunities for developing economies like Nigeria to understand the impact of energy policies.
Mishiwo <i>et al.</i> (2023),	Infant mortality rate and economic growth in Nigeria: An empirical analysis.	Multiple Linear Regression (MLR) model.	results disclosed that infant mortality has a negative significant influence on economic growth which translates to a	The gap observed in this research is the inclusion of more variables for robust dynamics of the relationship

Author(s) & Year	Topic	Methodology		Findings	Gaps
				negative relationship implying that an improved economic growth in Nigeria would result in a drastic decline in infant mortality.	and opposite analysis among the variables.
Guo <i>et al.</i> (2023)	The asymmetric impact of economic uncertainty on human health in China's economy.	Quantile Regressive Distributive (QARDL) Technique.	Auto Lag	The relationship as observed among the negative shock of economic uncertainty and infant mortality shows a negative significance at the highest quantiles.	Economic volatility is known to have a lingering effect that tends to reoccur in the future. This creates an avenue for further investigation of these extensive impacts.
Guo <i>et al.</i> (2024)	An investigation of the intensifying threat to public health caused by increased CO2 emissions from energy and modern urbanization in SAARC countries.	Panel Regressive Distributed Moments Quantile Regressions.	Auto Lag Method of	The result an impressive positive impact in the case of infant mortality, the study emphasized that renewable energy, urbanization, GDP, and industrialization negatively affect infant mortality, while CO2 emissions positively impact infant mortality in both short and long-run scenarios.	This provides a new perspective on other factors that may influence infant mortality in west Africa especially in Nigeria with potential shift in the use of green energy.
Li and Li (2024)	Government health expenditures and neonatal mortality in China: implication for sustainable development goal 3.	Canonical Cointegrating Regression Technique		It was found that China has been able to meet SDG 3 as a result of suspected increased government health expenditure.	The gap lies on ascertaining Nigeria's progress towards achieving the sustainable development goals.
Alawiye (2024)	The impact of digital technology on	Thematical Analysis Method		This study found that there is a significant digital technological	There is avenue for investigating narrowly into the

Author(s) & Year	Topic	Methodology	Findings	Gaps
	healthcare delivery and patient outcomes.		impact on patient due to enhanced access, operational efficiency, and diagnostic accuracy.	role of technology on infant mortality rate.
Miftahu and Kawu (2024)	Impact of healthcare investment on infant mortality in Nigeria.	Autoregressive Distributed Lag Model (ARDL) Bound test technique	The study discovered a long-run relationship between healthcare expenditure and infant mortality in Nigeria.	The lack of enough variable may account less for the variations of infant mortality rates in Nigeria. This gives room for inclusion of relevant health related variables that influence the infant mortality rates.

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Null Hypothesis: IMR has a unit root		
Exogenous: Constant		
Lag Length: 1 (Automatic - based on SIC, max lag=8)		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.869888	0.3418
Test critical values:	1% level	-3.646342
	5% level	-2.954021
	10% level	-2.615817

Dependent Variable: D(IMR)
Method: Least Squares
Date: 07/23/25 Time: 20:21
Sample (adjusted): 1992 2024
Included observations: 33 after adjustments

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GDPPC UNIT ROOT TEST

Null Hypothesis: D(GDPPC) has a unit root

Exogenous: Constant

Lag Length: 0 (Automatic - based on SIC, max lag=8)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-4.161936	0.0027
Test critical values: 1% level	-3.646342	
5% level	-2.954021	
10% level	-2.615817	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation

Dependent Variable: D(GDPPC,2)

Method: Least Squares

Date: 07/23/25 Time: 20:44

Sample (adjusted): 1992 2024

Included observations: 33 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D (GDPPC (-1))	-0.964445	0.231730	-4.161936	0.0002
C	6.041746	82.98880	0.072802	0.9424
R-squared	0.358466	Mean dependent var		-50.42212
Adjusted R-squared	0.337772	S.D. dependent var		577.9503
S.E. of regression	470.3210	Akaike info criterion		15.20340
Sum squared resid	6857256.	Schwarz criterion		15.29410
Log likelihood	-248.8561	Hannan-Quinn criteria.		15.23392
F-statistic	17.32171	Durbin-Watson stat		1.604810
Prob(F-statistic)	0.000232			

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CHE UNIT ROOT TEST				
Null Hypothesis: D(CHE) has a unit root				
Exogenous: Constant, Linear Trend				
Lag Length: 6 (Automatic - based on SIC, max lag=8)				
		t-Statistic	Prob.*	
Augmented Dickey-Fuller test statistic		-3.905995	0.0258	
Test critical values:	1% level	-4.339330		
	5% level	-3.587527		
	10% level	-3.229230		
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(CHE,2)				
Method: Least Squares				
Date: 07/23/25 Time: 21:04				
Sample (adjusted): 1998 2024				
Included observations: 27 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D (CHE (-1))	-7.607170	1.947563	-3.905995	0.0010
D (CHE (-1),2)	6.035140	1.839585	3.280708	0.0042
D (CHE (-2),2)	4.959184	1.660248	2.987015	0.0079
D (CHE (-3),2)	3.756315	1.441461	2.605908	0.0179
D (CHE (-4),2)	2.911876	1.072882	2.714069	0.0142
D (CHE (-5),2)	1.437833	0.690934	2.080997	0.0520
D (CHE (-6),2)	0.988587	0.392416	2.519230	0.0214
C	-41.10915	27.46130	-1.496985	0.1517
@TREND ("1990")	6.465026	2.142135	3.018029	0.0074
R-squared	0.888230	Mean dependent var		1.105185
Adjusted R-squared	0.838555	S.D. dependent var		97.50011
S.E. of regression	39.17577	Akaike info criterion		10.43520
Sum squared resid	27625.33	Schwarz criterion		10.86714
Log likelihood	-131.8751	Hannan-Quinn criteria		10.56364
F-statistic	17.88068	Durbin-Watson stat		1.915024
Prob(F-statistic)	0.000000			

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ACE UNIT ROOT TEST				
Null Hypothesis: D(ACE) has a unit root				
Exogenous: Constant				
Lag Length: 0 (Automatic - based on SIC, max lag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-9.264986	0.0000
Test critical values:	1% level		-3.646342	
	5% level		-2.954021	
	10% level		-2.615817	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(ACE,2)				
Method: Least Squares				
Date: 07/23/25 Time: 21:13				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D (ACE (-1))	-1.488263	0.160633	-9.264986	0.0000
C	1.555216	0.536151	2.900705	0.0068
R-squared	0.734680	Mean dependent var		0.026364
Adjusted R-squared	0.726121	S.D. dependent var		5.599575
S.E. of regression	2.930450	Akaike info criterion		5.046881
Sum squared resid	266.2137	Schwarz criterion		5.137578
Log likelihood	-81.27353	Hannan-Quinn criteria		5.077398
F-statistic	85.83996	Durbin-Watson stat		2.094962
Prob(F-statistic)	0.000000			

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AID UNIT ROOT TEST				
Null Hypothesis: D(AID) has a unit root				
Exogenous: None				
Lag Length: 0 (Automatic - based on SIC, max lag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-2.226164	0.0271
Test critical values:	1% level		-2.636901	
	5% level		-1.951332	
	10% level		-1.610747	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(AID,2)				
Method: Least Squares				
Date: 07/24/25 Time: 15:38				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D (AID (-1))	-1.295054	0.581742	-2.226164	0.0332
R-squared	0.112622	Mean dependent var		-16.54545
Adjusted R-squared	0.112622	S.D. dependent var		106.6818
S.E. of regression	100.4950	Akaike info criterion		12.08793
Sum squared resid	323176.0	Schwarz criterion		12.13328
Log likelihood	-198.4508	Hannan-Quinn criteria		12.10319
Durbin-Watson stat	1.123641			

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APPENDIX C: Correlation Analysis

	IMR	GDPPC	CHE	ACE	AID
IMR	1.000000	-0.6998	-0.9294	-0.9343	-0.8282
GDPPC	-0.6998	1.000000	0.5877	0.5970	0.8437
CHE	-0.9294	0.5877	1.000000	0.8613	0.6478
ACE	-0.9343	0.5970	0.8613	1.000000	0.7114
AID	-0.8282	0.8437	0.6478	0.7114	1.000000

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APPENDIX D: ARDL Result

ARDL Cointegrating And Long Run Form

Dependent Variable: IMR

Selected Model: ARDL (1, 4, 4, 4, 1)

Date: 07/28/25 Time: 17:02

Sample (adjusted): 1994 2024

Included observations: 31 after adjustments

Cointegrating Form

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
IMR (-1)	0.893954	0.044039	20.29891	0.0000
GDPPC	0.000592	0.000142	4.159451	0.0013
GDPPC (-1)	-5.93E-05	0.000170	-0.349357	0.7329
GDPPC (-2)	1.14E-05	0.000155	0.073032	0.9430
GDPPC (-3)	4.93E-05	0.000168	0.293812	0.7739
GDPPC (-4)	-0.000328	0.000133	-2.472568	0.0294
CHE	-0.001133	0.001284	-0.882609	0.3948
CHE (-1)	0.001448	0.001486	0.974765	0.3489
CHE (-2)	-0.001865	0.001669	-1.117510	0.2857
CHE (-3)	-0.001399	0.001690	-0.827523	0.4241
CHE (-4)	0.004789	0.001403	3.413318	0.0051
ACE	-0.040936	0.017438	-2.347477	0.0369
ACE (-1)	-0.077347	0.019766	-3.913185	0.0021
ACE (-2)	-0.046867	0.021770	-2.152840	0.0524
ACE (-3)	-0.014440	0.021490	-0.671929	0.5144
ACE (-4)	-0.037625	0.017690	-2.126892	0.0549
AID	-0.002832	0.000841	-3.365440	0.0056
AID (-1)	-0.003434	0.001667	-2.059901	0.0618
CointEq(-1)*	-0.106046	0.005756	-18.42369	0.0000
CointEq = IMR - (0.0025*GDPPC + 0.0174*CHE -2.0483*ACE -0.0591*AID)				

Long Run Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GDPPC	0.002508	0.001386	1.808857	0.0956
CHE	0.017357	0.045567	0.380902	0.7099
ACE	-2.048302	0.512111	-3.999721	0.0018
AID	-0.059087	0.013541	-4.363389	0.0009
C	19.58498	7.335518	2.669883	0.0204

ARDL Bounds Test

Date: 07/28/25 Time: 18:07

Sample: 1990 2024

Included observations: 31

Null Hypothesis: No long-run relationships exist

Test Statistic	Value	Significance	I(0) Asymptotic: n=1000	I(1)
F-statistic	50.91483	10%	2.45	3.52
k	4	5%	2.86	4.01
		2.5%	3.25	4.49
		1%	3.74	5.06

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Test Equation:

Dependent Variable: D(IMR)

Methods: Least Squares

Case 3: Unrestricted Constant and No Trend

Date: 07/28/25 Time: 18:07

Sample: 1990 2024

Included observations: 31

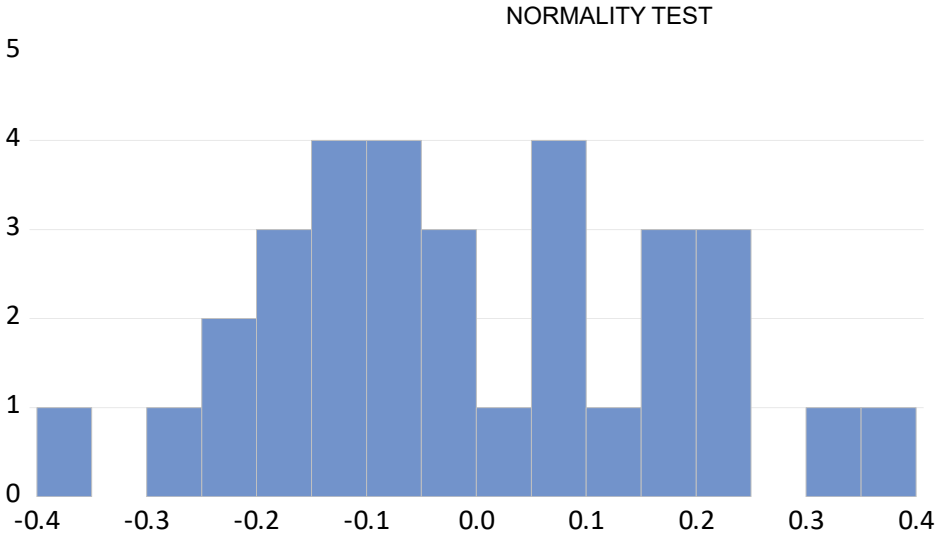
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	19.58498	7.335518	2.669883	0.0204
IMR(-1)*	-0.106046	0.044039	-2.407982	0.0330
GDPPC(-1)	0.000266	0.000152	1.753162	0.1051
CHE(-1)	0.001841	0.004091	0.449892	0.6608
ACE(-1)	-0.217215	0.057377	-3.785748	0.0026
AID(-1)	-0.006266	0.001797	-3.486229	0.0045
D(GDPPC)	0.000592	0.000142	4.159451	0.0013
D(GDPPC(-1))	0.000267	0.000111	2.397249	0.0337
D(GDPPC(-2))	0.000278	0.000117	2.370995	0.0353
D(GDPPC(-3))	0.000328	0.000133	2.472568	0.0294
D(CHE)	-0.001133	0.001284	-0.882609	0.3948
D(CHE(-1))	-0.001526	0.002513	-0.607160	0.5551
D(CHE(-2))	-0.003391	0.001778	-1.906998	0.0807
D(CHE(-3))	-0.004789	0.001403	-3.413318	0.0051
D(ACE)	-0.040936	0.017438	-2.347477	0.0369
D(ACE(-1))	0.098932	0.044278	2.234309	0.0453
D(ACE(-2))	0.052064	0.031462	1.654833	0.1239
D(ACE(-3))	0.037625	0.017690	2.126892	0.0549
D(AID)	-0.002832	0.000841	-3.365440	0.0056
R-squared	0.999975	Mean dependent var	87.56790	
Adjusted R-squared	0.999937	S.D. dependent var	23.62261	
S.E. of regression	0.187949	Akaike info criterion	-0.228569	
Sum squared resid	0.423897	Schwarz criterion	0.650326	
Log likelihood	22.54282	Hannan-Quinn criter.	0.057929	
F-statistic	26327.80	Durbin-Watson stat	2.949797	
Prob(F-statistic)	0.000000			

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APPENDIX E: POST ESTIMATION TESTS



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SERIAL CORRELATION TEST

Breusch-Godfrey Serial Correlation LM Test:

Null hypothesis: No serial correlation at up to 2 lags

F-statistic	0.767301	Prob. F (2,15)	0.4816
Obs*R-squared	2.969971	Prob. Chi-Square (2)	0.2265

Test Equation:

Dependent Variable: RESID

Method: ARDL

Date: 07/25/25 Time: 19:44

Sample: 1993 2024

Included observations: 32

Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
IMR(-1)	0.007056	0.066232	0.106534	0.9166
GDPPC	8.30E-05	0.000200	0.414966	0.6840
GDPPC(-1)	-0.000182	0.000266	-0.683561	0.5047
GDPPC(-2)	0.000115	0.000197	0.584866	0.5673
CHE	0.000360	0.001319	0.273316	0.7883
ACE	0.013543	0.034117	0.396939	0.6970
ACE(-1)	-0.015536	0.030573	-0.508178	0.6187
ACE(-2)	-0.007031	0.027464	-0.255993	0.8014
ACE(-3)	-0.004038	0.023593	-0.171157	0.8664
AID	7.68E-05	0.000767	0.100142	0.9216
AID(-1)	0.000390	0.003316	0.117645	0.9079
AID(-2)	0.000578	0.005019	0.115065	0.9099
AID(-3)	-0.000814	0.003799	-0.214383	0.8331
C	-0.544013	7.826868	-0.069506	0.9455
@TREND	0.018229	0.172907	0.105427	0.9174
RESID(-1)	0.018784	0.363057	0.051739	0.9594
RESID(-2)	-0.452420	0.371077	-1.219205	0.2416
R-squared	0.092812	Mean dependent var		-4.16E-15
Adjusted R-squared	-0.874856	S.D. dependent var		0.179961
S.E. of regression	0.246413	Akaike info criterion		0.341198
Sum squared resid	0.910790	Schwarz criterion		1.119870
Log likelihood	11.54083	Hannan-Quinn criter.		0.599306
F-statistic	0.095913	Durbin-Watson stat		2.033011
Prob(F-statistic)	0.999986			

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HETEROSKEDASTICITY TEST

Heteroskedasticity Test: Breusch-Pagan-Godfrey			
Null hypothesis: Homoskedasticity			
F-statistic	0.718693	Prob. F(14,17)	0.7309
Obs*R-squared	11.89779	Prob. Chi-Square(14)	0.6145
Scaled explained SS	2.329355	Prob. Chi-Square(14)	0.9998

Test Equation:
 Dependent Variable: RESID^2
 Method: Least Squares
 Date: 07/25/25 Time: 20:29
 Sample: 1993 2024
 Included observations: 32

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.106954	1.245618	-0.085864	0.9326
IMR(-1)	0.002836	0.010072	0.281564	0.7817
GDPPC	-2.70E-05	2.55E-05	-1.060284	0.3038
GDPPC(-1)	1.51E-05	3.29E-05	0.457413	0.6532
GDPPC(-2)	-2.11E-05	2.69E-05	-0.783178	0.4443
CHE	-5.36E-05	0.000210	-0.255494	0.8014
ACE	-0.006620	0.004201	-1.575917	0.1335
ACE(-1)	-0.002096	0.004381	-0.478488	0.6384
ACE(-2)	0.001820	0.004161	0.437306	0.6674
ACE(-3)	0.000493	0.003615	0.136353	0.8931
AID	-2.36E-05	0.000125	-0.189544	0.8519
AID(-1)	1.64E-05	0.000459	0.035597	0.9720
AID(-2)	0.000138	0.000723	0.190899	0.8509
AID(-3)	0.000240	0.000554	0.433369	0.6702
@TREND	0.007587	0.025207	0.300997	0.7671
R-squared	0.371806	Mean dependent var		0.031374
Adjusted R-squared	-0.145530	S.D. dependent var		0.037546
S.E. of regression	0.040185	Akaike info criterion		-3.285646
Sum squared resid	0.027453	Schwarz criterion		-2.598582
Log likelihood	67.57033	Hannan-Quinn criter.		-3.057904
F-statistic	0.718693	Durbin-Watson stat		1.722893
Prob(F-statistic)	0.730875			

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APPENDIX F: Data for Regression

4.1 Data Presentation

Table 4.1: Data for Regression

Year	IMR (per 1k deaths)	GDPPC (\$)	CHE(Billions)	ACE (%)	AID (\$)
1990	125.865	568	0.5	27.3	50
1991	125.87	609	0.62	35.23	52
1992	125.874	520	0.15	36.1	55
1993	125.86	552	3.87	36.96	58
1994	124.39	762	2.09	37.83	60
1995	122.92	1303	3.32	38.69	75
1996	121.45	1674	3.02	39.55	80
1997	119.98	1765	3.89	40.4	90
1998	118.5	1872	4.74	41.25	100
1999	115.59	494	16.64	44.9	120
2000	112.68	563	15.22	43.18	150
2001	109.78	583	24.52	43.91	180
2002	106.87	734	40.62	44.63	210
2003	103.96	787	33.27	52.2	250
2004	101.03	993	34.2	46.08	300
2005	98.106	1250	55.66	46.82	400
2006	95.18	1652	62.25	47.58	430
2007	92.255	1876	81.91	50.13	460
2008	89.33	2228	98.22	50.3	490
2009	86.458	1884	90.2	49.97	510
2010	83.587	2280	99.1	48	550
2011	80.715	2505	231.8	55.9	570
2012	77.844	2728	197.9	53.02	575
2013	74.972	2977	179.99	55.6	580
2014	72.406	3201	195.98	54.15	590
2015	69.84	2680	257.7	52.5	600
2016	67.708	2145	200.82	59.3	580
2017	64.708	1942	245.19	54.4	570
2018	62.142	2126	296.44	56.5	550
2019	60.662	2334	388.37	56	540
2020	59.181	2075	369.35	55.4	500
2021	57.701	2066	351.39	59.5	520
2022	56.22	2163	207.39	60.1	550
2023	54.74	2500	404.08	61.2	604
2024	53.7	877.07	434.79	70	60

Source: CBN Statistical Bulletin 2023 and World Bank Online Databank 2024.

EMPIRICAL ANALYSIS OF THE IMPACT OF SELECTED MACROECONOMIC VARIABLES ON INFANT MORTALITY IN NIGERIA (1990-2024).

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